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Generative Risk-neutral Option Market Models

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Abstract

We present a statically and dynamically arbitrage-free European option market model with smooth interpolation between option prices, based on the SANOS Buehler et al. [2026b] framework, and prove that its dynamics admit a unique strong solution.

In spirit, it is a stochastic model for local volatility when the latter is used as a representation of a European option price surface, not as a pricing model.

We illustrate the strength of our approach on S&P Index Option data from January 2022 to August 2025. We show that the model's initial state fits within bid/ask for more than 95% of relevant market option prices.[1] We then train the model dynamics under $\mathbb{P}$ and $\mathbb{Q}$ and show that the model is indeed free of arbitrage while fitting the dynamics of the market well. We also propose a range of BHW-tests to test for absence of arbitrage.

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[1] The model presented was trained for daily expiries up to 25 business days and strikes between $[\exp(-2.5\sigma_{atm}\sqrt{t}), \exp(+1.5\sigma_{atm}\sqrt{t})]$.

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1 Introduction

This article presents an arbitrage-free market model for smooth surfaces of European option prices. While not strictly the first of its kind, it is the first model with an industry-grade ability to fit the current market, allow for smooth interpolation of the entire option surface at each time, and provide an explicit construction of the continuous time risk-neutral drift for both (discrete local) volatilities and spot. It also has a natural estimation strategy for its parameter space.

The core architecture follows the SANOS approach Buehler et al. [2026b] which models the stock price as the product of a time-and-state discrete process X and a smoothing process Y, in our case Black & Scholes. The former is used to encode the option market using a discrete form of local volatilities as code book. That does not mean that the model is a local volatility model – we merely use the implied local volatilities as a convenient representation of the option surface. Compared to using implied volatilities the advantage is that mere positivity of the local volatilities is sufficient to ensure the option market is free of static arbitrage at all times. We have shown in Buehler et al. [2026b] that this representation can be fitted to real life market data very efficiently using linear programming.

The key contribution in this paper is showing how to construct a dynamic model of SANOS option call prices which are not only statically but also dynamically free of arbitrage. We show the theoretical construction of the corresponding risk-neutral measure $\mathbb{Q}$ and prove that its dynamics admit a unique strong solution. We then discuss implementation and estimate our model on S&P Index Option data from January 2022 to August 2025. We fit our daily model parameters to observed bid/ask prices and then proceed to estimate both $\mathbb{Q}$ and its drift under $\mathbb{P}$. We show that on this data set more than 95% of calibrated model prices fit within observed market bid/ask spreads even with the simplistic grid architecture we present here.

We then show that model dynamics under the risk-neutral indeed have numerically zero conditional and unconditional drift. We test this first with a restricted universe of 570 tradable instruments in a number of different set ups. We then also show that even using all 5001 instruments the risk-neutrality is preserved. Ours is the first model being able to show such a result for a number of instruments which far exceeds the number of driving stochastic factors.

The model presented here is distinct from our own DYSANOS Buehler et al. [2026a]. There, we focused on learning realistic market dynamics for smooth statically arbitrage-free SANOS surfaces under $\mathbb{P}$. We did also construct an equivalent risk-neutral measure $\mathbb{Q}$ numerically using the *martingale measure* approach introduced in Buehler et al. [2022]. While this method is

model-free and can in theory be applied to any option market model under $\mathbb{P}$, it is numerically heavy and in practice restricted to making only a fairly limited number of the overall tradable instruments dynamically arbitrage-free.[2] In the current work we cover all options with an explicit construction of the risk-neutral drift, which is far more efficient than the numerical martingale measure approach.

Related Work

Our approach is most closely related to the seminal but unpublished work Wissel [2007] which extends their earlier works Schweizer and Wissel [2008a] and Schweizer and Wissel [2008b]. From a theoretical perspective, our approach is a version of Carmona and Nadtochiy [2009] where the authors derive related risk-neutral drift conditions for continuous time local volatility models, but without a tangible implementation strategy. Our results can be seen as a joint extension of the two papers, plus smoothing and market-fit from our SANOS Buehler et al. [2026b].

In Carmona and Nadtochiy [2012] the authors solve the problem of finding a dynamically arbitrage-free evolution of the full option surface via products of independent Lévy processes. While structurally very elegant, option prices generated by products of independent increment processes do not typically fit well to equity index European option surfaces. Our approach also has the advantage of being numerically accessible. In particular, our model prices can be fit very efficiently within nearly all[3] of observed market bid/ask prices using the linear programming SANOS Buehler et al. [2026b] calibration method.

There are a number of works on generative equity option market dynamics – not least our own 2019 work Wiese et al. [2019] and the multi-asset version Wiese et al. [2021] – which simulate discrete strictly arbitrage-free option surfaces. However, most other approaches are not strictly free of arbitrage and do not have analytical corresponding risk-neutral dynamics. To address the absence of an analytical risk-neutral measure, we presented in Buehler et al. [2022] a model-free approach to learning such a measure. However, that method is numerically challenging for a large number of tradable options as discussed in Buehler et al. [2026a].

For interest rate option markets the reference model in the same class is the BGM Model Brace et al. [1997].

2 Stochastic Smooth Discrete Local Volatility

In this section we introduce the core model architecture for our *Stochastic Smooth Discrete Local Volatility model*. We follow the core SANOS Buehler et al. [2026b] idea of writing our equity process S as product

$$S_t = X_t Y_t \tag{1}$$

of a *Stochastic Discrete Local Volatility (SDLV)* process X which is used to fit to current market conditions and whose state parameters will have consistent $\mathbb{P}$ and $\mathbb{Q}$ dynamics, and an independent[4] and rather trivial smoothing process Y whose dynamics under $\mathbb{P}$ and $\mathbb{Q}$ are the same.

The process Y will just be a Black & Scholes process with fixed volatility 5%. Given observed bid/ask market data it was shown in Buehler et al. [2026b] how to fit a discrete local volatility surface in the presence of such process.

[2] In Buehler et al. [2026a] we used 49 instruments per simulated date.

[3] In below experiments, above 95%.

[4] Under both $\mathbb{P}$ and $\mathbb{Q}$.

In the remainder of this section we focus on the SDLV process itself.

2.1 Stochastic Discrete Local Volatility (SDLV)

Our *Stochastic Discrete Local Volatility (SDLV)* risk-neutral model will be defined on a *fixed* expiry and cash strike grid. It will not use a floating grid in its option market dynamics.[5] We will use "t" to refer to model time. Denote initial spot by $X_0 > 0$.

For now let $0 = T_0 < T_1 < \cdots < T_M$ be the fixed model expiry dates, and assume that $K^0 < K^1 < \cdots < X_0 < \cdots < K^{N+1}$ are the fixed model cash strikes, where K^0 and K^{N+1} are boundary strikes which are far away from the inner strikes, cf. Buehler et al. [2026b].[6]

Assume that t is the current model time and let T_k be the front expiry such that $T_{k-1} \le t < T_k$. Our model is given at any time t by the current spot X_t and future densities q_t^j, where $q_t^{j,i}$ represents the risk-neutral density of the i-th cash strike at future expiry T_j.

Such discrete densities have been shown in Buehler and Ryskin [2015] to admit an equivalent representation via (squared) *discrete local volatilities* $\Sigma_t^{j,1}, \ldots, \Sigma_t^{j,N}$ in the following form: there is a tridiagonal matrix Ω_j[7] for each $j > k$ such that

$$q_t^{j-1} = M_t^j q_t^j \ , \qquad M_t^j := I - \Omega_j S_t^j \ , \qquad S_t^j := \operatorname{diag}(0, \Sigma_t^{j,1}, \ldots, \Sigma_t^{j,N}, 0) \tag{2}$$

where M_t^j is an invertible tridiagonal M-matrix for every non-negative S_t^j. The key point is that q_t^j for $j > k$ are uniquely determined martingale densities for any given initial density q_t^k and the surface of non-negative (squared) discrete local vols.

2.2 Risk-Neutral Dynamics

Under a risk-neutral measure $\mathbb{Q}$ densities are martingales. We will now turn this property into conditions on dynamics of the (squared) discrete local volatilities, and, jointly the front density and spot. Our approach follows in spirit that of Wissel [2007] in the sense that we rely on the martingale property of a discrete set of tradable derivative assets. However, we reframe it in a much more accessible coordinate system.

Assume that W is a d-dimensional Brownian motion. We designate W^1 as the spot factor: spot is always driven by dW_t^1.

2.2.1 Future Densities

Each density coordinate is a martingale. We therefore specify the front-density diffusion as

$$dq_t^k = E_t^{k;1}\, dW_t^1 \ , \qquad E_t^{k;1} := \operatorname{diag}(q_t^k)\eta_t^{k;1} \ , \qquad E_t^{k;f} := 0 \quad (f = 2, \ldots, d),$$

[5] This is not to be confused with how the model parameters are estimated using historic market data. We explain the estimation in section 3.

[6] In our experiments the inner strikes are given as range of strikes in from $K^1 = \exp(-2.5\sigma_{atm}(T_M)\sqrt{T_M})$ to $K^N = \exp(+1.5\sigma_{atm}(T_M)\sqrt{T_M})$ with boundaries set to $K^0 = K^1/2$ and $K^{N+1} = 3K^N/2$; see section 3.1 for specifics.

[7] The matrix $\Omega_j \in \mathbb{R}^{(N+2)\times(N+2)}$ is given as follows: let $\gamma^{\pm i} := \pm(K^i)^2(T_j - T_{j-1})/(K^{i+1} - K^{i-1})/(K^{i\pm1} - K^i)$ for $i = 1, \ldots, N$. Then Ω_j is the tridiagonal matrix whose i-th column, $i = 1, \ldots, N$, has the entries

$$(\Omega_j)_{i-1,i} = \gamma^{-i}, \quad (\Omega_j)_{i,i} = -(\gamma^{-i} + \gamma^{+i}), \quad (\Omega_j)_{i+1,i} = \gamma^{+i},$$

and all other entries zero, so that $\mathbf{1}^\top \Omega_j = 0$ and $K^\top \Omega_j = 0$; cf. Buehler et al. [2026b].

where $E_t^{k;1} \in \mathbb{R}^{N+2}$ must satisfy $\mathbf{1}^\top E_t^{k;1} = 0$, so that q_t^k retains unit mass. The remaining Brownian factors do not directly move the front density. They drive the stochastic DLV surface, while $\xi_t^{j,\cdot;1}$ determines their correlation with spot. Future density volatilities η_t^j for $j > k$ will be obtained recursively.

For future expiries T_j with $j > k$, assume the discrete local volatility squared is given as

$$\frac{d\Sigma_t^j}{\Sigma_t^j} = \mu_t^j \, dt + \xi_t^j \, dW_t, \tag{3}$$

where $\xi_t^j \in \mathbb{R}^{N\times d}$ is given and $\mu_t^j \in \mathbb{R}^N$ is the unknown risk-neutral drift. We write this as

$$dS_t^j = A_t^j \, dt + \sum_{f=1}^{d} B_t^{j;f} \, dW_t^f,$$

with $A_t^j := \operatorname{diag}(0, \Sigma_t^{j,1}\mu_t^{j,1}, \ldots, \Sigma_t^{j,N}\mu_t^{j,N}, 0)$ and $B_t^{j;f} := \operatorname{diag}(0, \Sigma_t^{j,1}\xi_t^{j,1;f}, \ldots, \Sigma_t^{j,N}\xi_t^{j,N;f}, 0)$. With the above

$$dM_t^j = -\Omega_j \, dS_t^j = -\Omega_j A_t^j \, dt - \sum_{f=1}^{d} \Omega_j B_t^{j;f} \, dW_t^f.$$

Setting $dq_t^j = \sum_f E_t^{j;f} \, dW_t^f$ for an unknown $E_t^{j;f}$ yields

$$\begin{aligned} dq_t^{j-1} &= M_t^j \, dq_t^j + (dM_t^j) q_t^j + d[M^j, q^j]_t & (4)\\ &= \sum_{f=1}^{d} \left(M_t^j E_t^{j;f} - \Omega_j B_t^{j;f} q_t^j \right) dW_t^f - \Omega_j \left(A_t^j q_t^j + \sum_{f=1}^{d} B_t^{j;f} E_t^{j;f} \right) dt \ . & (5) \end{aligned}$$

Matching the diffusion coefficients gives

$$E_t^{j;f} = (M_t^j)^{-1} \left(E_t^{j-1;f} + \Omega_j B_t^{j;f} q_t^j \right) \in \mathbb{R}^{N+2} \quad \text{and} \quad \eta_t^{j,i;f} = \frac{E_t^{j,i;f}}{q_t^{j,i}} \ . \tag{6}$$

The drift must vanish, hence

$$\Omega_j \left(A_t^j q_t^j + \sum_{f=1}^{d} B_t^{j;f} E_t^{j;f} \right) = 0.$$

The restriction of Ω_j to vectors supported on the interior nodes is injective. Therefore the bracket vanishes node by node, giving

$$\mu_t^{j,i} = -\sum_{f=1}^{d} \xi_t^{j,i;f} \frac{E_t^{j,i;f}}{q_t^{j,i}} = -\sum_{f=1}^{d} \xi_t^{j,i;f} \eta_t^{j,i;f}, \qquad i = 1, \ldots, N. \tag{7}$$

Thus the prescribed front-density volatility and the prescribed DLV volatilities determine all future density volatilities and the risk-neutral DLV drifts by forward induction over the expiries.

2.2.2 Front Density and Spot

The remaining task is to specify both spot dynamics consistent with the front density and the respective front density volatilities.

Case $t = T_k - dt$

Assume we are at $t = T_k - dt$ and observe the front density q_t^k. Current spot is given as $X_t = K^\top q_t^k$. In order to sample a spot consistent with the discrete set of strikes, we can either draw a discrete sample or fit some other distribution exactly to the observed option prices (fitting to densities or probabilities will not work in general).

The simplest method is to draw spot from this discrete distribution using the spot factor W^1. Let F_t^k denote the conditional CDF which assigns probability $q_t^{k,i}$ to strike K^i, and let Φ denote the standard normal CDF. Then

$$X_{T_k} = (F_t^k)^{-1}(\Phi(Z)), \qquad Z := \frac{W_{T_k}^1 - W_t^1}{\sqrt{dt}} \sim N(0,1).$$

Correlation between spot and a future DLV coordinate is specified by that coordinate's loading $\xi_t^{j,i;1}$ on W^1.

Conditional on this definition of terminal spot, the front-density diffusion is exact if the Brownian increment is revealed progressively: define

$$Q_i := \sum_{\ell=0}^{i} q_t^{k,\ell}, \qquad a_i := \Phi^{-1}(Q_i), \qquad a_{-1} := -\infty, \quad a_{N+1} := +\infty,$$

and let ϕ be the standard normal density. For $t \le s < T_k$, set

$$z_i(s) := \frac{\sqrt{dt}\, a_i - (W_s^1 - W_t^1)}{\sqrt{T_k - s}}.$$

The time-s front density is the conditional probability

$$q_s^{k,i} := \mathbb{P}\big(X_{T_k} = K^i \mid \mathcal{F}_s\big) = \Phi(z_i(s)) - \Phi(z_{i-1}(s)). \tag{8}$$

It is therefore a martingale. Applying Itô's formula gives the exact diffusion

$$dq_s^{k,i} = E_s^{k,i;1}\, dW_s^1, \qquad E_s^{k,i;1} = \frac{\phi(z_{i-1}(s)) - \phi(z_i(s))}{\sqrt{T_k - s}}, \qquad t \le s < T_k, \tag{9}$$

with zero drift. Since $z_i(t) = a_i$, its initial value is

$$E_t^{k,i;1} = \frac{\phi(a_{i-1}) - \phi(a_i)}{\sqrt{dt}} \quad \text{hence} \quad \eta_t^{k,i;1} = \frac{\phi(a_{i-1}) - \phi(a_i)}{q_t^{k,i}\sqrt{dt}} \quad \text{for } i = 0, \dots, N+1 \tag{10}$$

while $E_t^{k,i;f} = \eta_t^{k,i;f} = 0$ for $f = 2, \dots, d$. More generally, $\eta_s^{k,i;1} = E_s^{k,i;1}/q_s^{k,i}$ and the other factor loadings remain zero. Consequently,

$$X_s = K^\top q_s^k = \mathbb{E}[X_{T_k} \mid \mathcal{F}_s], \qquad dX_s = K^\top E_s^{k;1}\, dW_s^1,$$

so spot and the front density are driven by the same Brownian factor. At expiry, q_s^k converges to the point mass associated with the sampled strike. The displayed coefficient $E_t^{k,i;1}$ is only the initial value of the exact, state- and time-dependent coefficient $E_s^{k,i;1}$; holding it constant over the whole interval would instead be an Euler approximation. Likewise, sampling X_{T_k} all at once without progressively revealing W^1 produces a direct jump to the terminal point mass rather than a continuous density diffusion.

Case $t \ll T_k$

The same construction gives a natural exact transition over any step which does not cross the front expiry. Let

$$h := T_k - t, \qquad 0 < \delta \le h, \qquad \Delta W_t^1 := W_{t+\delta}^1 - W_t^1.$$

Starting from q_t^k, define Q_i and a_i as above. After drawing $\Delta W_t^1 \sim N(0, \delta)$, set

$$q_{t+\delta}^{k,i} = \Phi\left(\frac{\sqrt{h}\, a_i - \Delta W_t^1}{\sqrt{h-\delta}}\right) - \Phi\left(\frac{\sqrt{h}\, a_{i-1} - \Delta W_t^1}{\sqrt{h-\delta}}\right), \qquad \delta < h. \tag{11}$$

The corresponding spot is not sampled independently but is defined by

$$X_{t+\delta} := K^\top q_{t+\delta}^k = \mathbb{E}[X_{T_k} \mid \mathcal{F}_{t+\delta}]\,. \tag{12}$$

Consequently

$$\mathbb{E}_t[q_{t+\delta}^k] = q_t^k, \qquad \mathbb{E}_t[X_{t+\delta}] = X_t$$

exactly. In general $X_{t+\delta}$ is not a grid strike; only at T_k does the density collapse to a unit vector and spot become the corresponding strike.

This update is recursively consistent. Indeed, the Gaussian boundaries calculated from the cumulative probabilities of $q_{t+\delta}^k$ are precisely the residual boundaries appearing in (11). The construction may therefore be restarted from $q_{t+\delta}^k$ without changing the joint law. It is a simplex-valued information bridge: conditional on the exact terminal Gaussian variable, W^1 is a Brownian bridge; conditional only on the terminal strike, it is Brownian motion conditioned to terminate in an interval, or equivalently a mixture of Brownian bridges.

2.2.3 Existence

We have just described a set of risk-neutral model dynamics for a discrete process X_t whose future densities themselves are martingales driven by stochastic discrete local volatilities. Our construction is concise and lends itself to efficient numerical implementation and model parameter estimation.

With $L_t^{j,i} \coloneqq \log \Sigma_t^{j,i}$, (3) and the drift restriction (7), the Itô dynamics of the log of the DLV squared are governed by

$$dL_t^{j,i} = \left(-\sum_{f=1}^{d} \xi_t^{j,i;f} \eta_t^{j,i;f} - \frac{1}{2}\sum_{f=1}^{d} (\xi_t^{j,i;f})^2\right) dt + \sum_{f=1}^{d} \xi_t^{j,i;f}\, dW_t^f. \tag{13}$$

Thus, the only term that can blow up the drift is $\eta^{j,i;f} = E^{j,i;f}/q^{j,i}$, as the density entries at the wings become small. For the Gaussian bridge front (Section 2.2.2), η_t grows asymptotically at the rate $(T_k - t)^{-1}$ on strikes it abandons. Thus, we cannot derive bounds for Σ directly, and instead investigate the relationship of the density q and $R = \Sigma q$, which in turn will allow us to read off the DLV.

Throughout, fix k, T_k being the front, and let $B_t = W_t^1 - W_{T_{k-1}}^1$ be the Brownian motion centered at the origin. We assume that the front follows the Gaussian construction from Section 2.2.2 with mass distributed along interior strikes, i.e. $\sum_{i=1}^{N} q_{T_{k-1}}^{k,i} > 0$, and that ξ is locally Lipschitz in (B, L) and bounded by a deterministic function $c(t)$ that is square-integrable, i.e. $\|\xi\|_\infty \le c(t)$ with $\int_{T_{k-1}}^{T_k} c^2 < \infty$. The constant loadings that are estimated in Section 3 satisfy this. We will denote by J the node at which the spot settles at time T_k, i.e. $X_{T_k} = K^J$.

Theorem 2.1 (Existence)**.** *Let $L_{T_{k-1}} = \log \Sigma_{T_{k-1}}$ be finite.*

(i) There is a unique strong solution for L on $[T_{k-1}, T_k)$, and every $\Sigma^{j,i}$ stays finite and strictly positive.

(ii) If spot settles at an interior strike, $J \in \{1, \dots, N\}$, then L extends continuously to T_k with L_{T_k} finite.

(iii) If spot settles at an extreme strike $J \in \{0, N+1\}$, then $q^j_{T_k} = e_J$ for all $j > k$. Moreover, the remaining DLVs Σ no longer affect the model.

Lemma 2.2. *Let $\tau \le T_k$ be a stopping time and let U, V be two positive continuous local martingales defined on the interval $[T_{k-1}, \tau)$. Assume that U, V have relative volatilities u and $u + v$ respectively, i.e. $dU/U = u\,dW$ and $dV/V = (u+v)\,dW$. Following holds true:*

(a) The left limit $U_{\tau-}$ exists, and $U_{\tau-}$ is zero exactly when $\int_{T_{k-1}}^{\tau} \|u_t\|^2 dt = \infty$.

(b) If v is square integrable on the interval $[T_{k-1}, \tau)$, then $U_{\tau-} = 0$ if and only if $V_{\tau-} = 0$.

Proof. The idea is simple: U is an exponential, and dies if the volatility grows too fast.

(a) Write $U_t = U_{T_{k-1}} \exp(M_t - \frac{1}{2}\langle M\rangle_t)$ where $M_t = \int_{T_{k-1}}^t u_s\,dW_s$ and $\langle M\rangle_t = \int \|u_s\|^2 ds$. If the quadratic variation $\langle M\rangle_\tau$ is finite, then M converges to a finite limit, hence $U_{\tau-} > 0$. Otherwise $\langle M\rangle_\tau = \infty$, and by Dambis–Dubins–Schwarz $M_t = \beta_{\langle M\rangle_t}$ for some Brownian motion β run on the clock $s = \langle M\rangle_t$. By the strong law of large numbers, $\beta_s/s \to 0$, so $M_t - \frac{1}{2}\langle M\rangle_t = s(\beta_s/s - \frac{1}{2}) \to -\infty$ and $U_t \to U_{\tau-} = 0$.

(b) The same holds true for V, which has volatility $u + v$. Since $\int \|v\|^2 < \infty$, $\int \|u+v\|^2$ is finite if and only if $\int \|u\|^2$ is, so by (a) $U_{\tau-}$ and $V_{\tau-}$ are either both zero or both positive.

□

Proof of Theorem 2.1. (i) Since $\Sigma = e^L > 0$, every M^j is a non-singular M-matrix, the coefficients of (13) are locally Lipschitz and the densities are positive. Hence, there is a unique strong solution up to a maximal lifetime $\zeta \le T_k$ [Protter, 2005, Theorem V.38]. It remains to show $\zeta = T_k$.

Let $R^{j,i} := \Sigma^{j,i} q^{j,i}$ and recall that the density q^j is governed by $dq^{j,i} = q^{j,i}\eta^{j,i}\,dW$. We want to show that both processes fulfil the assumptions of Lemma 2.2. Applying the Itô product rule to $R^{j,i}$ we observe that it is a local martingale $dR^{j,i} = R^{j,i}(\eta^{j,i} + \xi^{j,i})\,dW$. Moreover, we can show that R^j is bounded by applying the call payoff matrix Π, where $\Pi_{l,m} = (K^m - K^l)^+$, to the code-book relation (2) expressed in terms of R^j

$$q^{j-1} = q^j - \Omega_j R^j \tag{14}$$

which gives $C^j - C^{j-1} = \Pi\Omega_j R^j$. Thus, $R^{j,i}$ is the calendar spread at K^i from T_{j-1} to T_j up to a positive constant. Since $X \in [K^0, K^{N+1}]$, call prices lie in $[0, K^{N+1}]$, so R^j must be bounded. Thus, both q^j and R^j are bounded martingales and therefore attain limits at ζ by the martingale convergence theorem. Apply Lemma 2.2 (b) with $\tau = \zeta$ and $v = \xi$, then both $q^{j,i}$ and $R^{j,i}$ can only reach zero together.

Next we show that a single zero-valued interior density node, determines that the all interior nodes must be zero-valued. Contrary, a single positive-valued interior node spreads to all interior nodes. Abbreviate $p = q^{j-1}$, $q = q^j$, $R = R^j$ at time ζ and write out row i of (14):

$$p_i = q_i + (\gamma^{-i} + \gamma^{+i})R_i - \gamma^{+(i-1)}R_{i-1} - \gamma^{-(i+1)}R_{i+1}.$$

From Lemma 2.2 (b) we know that $q_i = 0$ iff $R_i = 0$. Hence, if $q_i = 0$, then also $R_i = 0$ and $p_i = -\gamma^{+(i-1)}R_{i-1} - \gamma^{-(i+1)}R_{i+1}$. The left-side, p_i, is non-negative and the right-side non-positive. Therefore $p_i = 0$ and $R_{i\pm1} = 0$ and again by Lemma 2.2 (b) $q_{i\pm1} = 0$. This argument spreads across the grid such that all interior nodes have to be equal to zero. Mass can only be attained on an outer node. Conversely, if q^{j-1} has interior mass then the chain above cannot start and all $q^{j,i}$, and thus $R^{j,i}$, are positive.

Now, suppose that $\zeta < T_k$. Then the Brownian motion B_ζ is finite, and the front density q^k_ζ has still strictly positive interior entries. Applying the above inductive chain for $j = k+1, \dots, M$ we obtain that $\Sigma^{j,i}_{\zeta-} = R^{j,i}_{\zeta-}/q^{j,i}_{\zeta-} \in (0, \infty)$. Hence, L has a finite limit at ζ which means it could be extended to ζ, contradicting the maximality of ζ. Therefore $\zeta = T_k$ demonstrating existence.

(ii) If $J \in \{1, \dots, N\}$, the terminal front has interior mass and by the above inductive chain $q^{j,i}_{T_k}, R^{j,i}_{T_k} > 0$ for all $j > k$. Hence, $\Sigma^{j,i}_{T_k} = R^{j,i}_{T_k}/q^{j,i}_{T_k} \in (0, \infty)$ for all $j > k$, and therefore the solution of L extends to L_{T_k}.

(iii) Since $K^\top \Omega_j = 0$, the code-book conserves the mean. Hence, for all $j > k$ we have $K^\top q^j_{T_k} = K^\top q^k_{T_k} = X_{T_k} = K^J$. A probability vector $q^k_{T_k}$ defined on $K^0, \dots, K^{N+1}$ with mean on the extreme end forces $q^j_{T_k} = e_J$, $J \in \{0, N+1\}$. Again recalling the code-book recursion $q^{j-1} = (I - \Omega_j S^j)q^j$ and that $S^j = \mathrm{diag}(0, \Sigma^{j,1}, \dots, \Sigma^{j,N}, 0)$ it follows that for an exterior density $q^j = e_J$, we have $S^j e_J = 0$. Thus, $e_J = M^j e_J$ for all Σ and by induction $q^j_{T_k} = (M^j)^{-1} q^{j-1}_{T_k} = (M^j)^{-1} e_J = e_J$ for all $j > k$. Hence all future densities and therefore also all call option prices, are independent of Σ.

□

Whenever the spot settles on an interior strike, the next front shifts $k \to k+1$ with density $(M^{k+1}_{T_k})^{-1} e_J$. The theorem then can be re-applied to the interval $[T_k, T_{k+1}]$. If the spot settles at an extreme strike, (iii) applies and the spot process is absorbed at this state. Since there are a finite amount of expirations, the existence theorem therefore gives the process on the full interval $[0, T_M]$.

Corollary 2.3. *Under the assumptions of Theorem 2.1, q^j is a true martingale on $[T_{k-1}, T_k]$ for every $j \geq k$.*

Proof. By construction the drift of q^j vanishes identically (the node-by-node cancellation in (7)), so q^j is a local martingale on the interval of existence furnished by Theorem 2.1. Each coordinate $q^{j,i}_t$ is a probability and therefore satisfies $0 \leq q^{j,i}_t \leq 1$, i.e. q^j is a bounded process. A bounded local martingale is uniformly integrable and closed by its terminal value, hence a true martingale; in particular $\mathbb{E}_t[q^j_s] = q^j_t$ for all $T_{k-1} \leq t \leq s$ in the interval, with no additional integrability condition required. □

2.3 Sampling

In this section we discuss the sampling procedure for the discrete local volatility model. The final model is a product of this process and a simple Black & Scholes process with, say, volatility of 5%. It serves as a SANOS Buehler et al. [2026b] smoother and ensures that the overall option price surface is smooth in strike.

On a time grid $t_n < t_{n+1}$ with $\delta_n = t_{n+1} - t_n$, one may proceed as follows.

First draw one Brownian vector $\Delta W_n \in \mathbb{R}^d$, using its first coordinate both in (11) and in every DLV coordinate with a non-zero loading $\xi^{j,i;1}$. This preserves the specified leverage. Update the front density by (11), and calculate spot from (12). This part of the update has no time-step bias, preserves positivity and unit mass, and is exactly martingale preserving.

For the DLV squared it is useful to work with $L_t^{j,i} \coloneqq \log \Sigma_t^{j,i}$, which follows (13).

Even the frozen-coefficient exponential update

$$\Sigma_{n+1}^{j,i} = \Sigma_n^{j,i} \exp\left\{ \left(\mu_n^{j,i} - \frac{1}{2} \sum_f (\xi_n^{j,i;f})^2 \right) \delta_n + \sum_f \xi_n^{j,i;f} \Delta W_n^f \right\} \tag{15}$$

is preferable to Euler integration of Σ: it cannot create a negative local variance. For state-dependent coefficients, a genuine Itô weak-order-two stochastic Runge–Kutta method in the log variables reduces the remaining time-step bias.

After updating Σ^j reconstruct the densities expiry by expiry by solving the tridiagonal systems

$$q_{n+1}^j = \left(I - \Omega_j S_{n+1}^j \right)^{-1} q_{n+1}^{j-1}, \qquad j = k+1, \dots, M. \tag{16}$$

This enforces the DLV relation at every simulated state. Because the matrix is an M-matrix for non-negative Σ, the solve preserves positivity.

The drift condition (7) removes the infinitesimal drift of every density coordinate. It does not, by itself, make a generic finite-step discretisation exactly unbiased, since the map $\Sigma^j \mapsto (M^j)^{-1} q^{j-1}$ is nonlinear. Thus (15), or any ordinary higher-order SDE scheme, retains a time-discretisation bias which should vanish under step refinement.

Our experiments in Section 4 use this sampling scheme with antithetically paired Brownian increments.

Remark 2.4 (Alternative discrete-branch construction). If exact martingality under a chosen finite-step simulation law is required, it can instead be imposed directly. Choose symmetric discrete shock branches $\Delta W_{n,b}$ with probabilities w_b and construct the front-density branches so that

$$\sum_b w_b q_{n+1,b}^k = q_n^k \ .$$

Parameterise each positive DLV update with an unknown drift correction $c_n^j \in \mathbb{R}^N$. For example, replace the log drift in (15) by $c_n^{j,i}$. For each expiry in turn, solve the nonlinear system

$$\sum_b w_b q_{n+1,b}^j(c_n^j) = q_n^j, \qquad q_{n+1,b}^j(c_n^j) = \left(I - \Omega_j S_{n+1,b}^j(c_n^j) \right)^{-1} q_{n+1,b}^{j-1}. \tag{17}$$

Only N coordinates are independent because mass and first moment are already preserved, matching the N drift corrections. Initialising the nonlinear solver with the continuous-time drift (7) gives a stable finite-step correction. Equation (17) makes the densities exactly martingales under the discrete branch law; symmetric or antithetic branches reduce variance but do not replace this condition.

3 Estimation

We briefly describe our estimation strategy. There are two components: fitting daily the "implied" discrete densities and therefore discrete local volatilities to observed bid/ask spreads, and then an estimation of the vol of those discrete local volatilities, and their statistical drift.

The former represents today's fit of the model to the market, while the latter captures the evolution of the model parameters over time.

The summary here is brief; refer to DYSANOS Buehler et al. [2026a] for a comprehensive discussion on parameter mapping and estimation considerations.

3.1 Data and state representation

We estimate a family of models which has 25 expiries at one-business-day intervals. The model is aimed at S&P 500 index options whose main trading activity occurs in the short-dated options segment.

Each daily state has 25 model expiries at one-business-day intervals and 200 interior strike nodes, together with two artificial boundary nodes. The 200 inner nodes are generated from fixed standardized log-moneyness templates on $[-2.5, 1.5]$.[8] They therefore share one strike support across the 25 expiries on a given date, while that support floats between dates with the observed ATM term structure. A fixed 5% annual Black volatility supplies the independent SANOS smoothing factor.

We use European S&P 500 index-option quotes via WRDS[9] from OptionMetrics IvyDB on 891 valid market dates from 3 January 2022 to 29 August 2025. After the cleaning and eligibility pipeline described in appendix B, the retained sample contains 2,192,458 original call and put bid–ask intervals with at most 25 business days to expiry. Prices and strikes are forward-normalized, while the contemporaneous spot is retained separately.

3.2 End-to-end fitting

Let $x_d \in \mathbb{R}^{25\times 200}$ denote the unconstrained state on date d. Its DLV-squared coordinates are decoded as

$$\Sigma_d^{j,i} = \min\{9, 10^{-4} + \exp(x_d^{j,i})\}.$$

The SANOS recursion then produces all densities and quote prices. Every retained market quote is priced at its own strike and expiry; quotes between two model expiries use the corresponding

[8] The expiry grid is fixed at $T_j = j/252$, $j = 1, \ldots, 25$. A normalized-moneyness coordinate z at expiry τ is mapped to the forward-relative strike

$$k(z,\tau) = \exp\left[z\sqrt{\frac{\sigma_\epsilon^2\epsilon + \sigma_{\mathrm{ATM}}(\tau)^2\tau}{1+\epsilon}}\right], \qquad \epsilon = \frac{1}{8\times 252},$$

where $\sigma_\epsilon = \sigma_{\mathrm{ATM}}(1/252)$ and $\sigma_{\mathrm{ATM}}(\tau)$ is the regularized annual ATM Black volatility. The cash strike is $K = F_0(\tau)k$; in the normalized simulations $F_0(\tau) = 1$. The strike construction uses 100 fixed coordinates for the one-day ATM scale and 100 for the 25-day ATM scale, both on $[-2.5, 1.5]$. On each calibration date the resulting log-relative-strike candidates are merged and sorted, and given a minimum gap equal to 5% of the average log-strike spacing. Exponentiation gives one 200-node relative-strike grid shared by all 25 expiries; the boundary strikes are one-half of the lowest and one and a half times the highest inner strike. Thus the expiry and normalized-moneyness templates are fixed, and the resulting cash strikes are fixed along each simulated path. For historical calibration the relative-strike grid floats between dates because the ATM scales change.

[9] `https://wrds-www.wharton.upenn.edu/`

linear time interpolation. The fit is therefore performed directly against the original market bid and ask, not against an intermediate linear-programming state.

The *direct* specification fits all 5,000 coordinates of x_d independently for every date. The *affine* specifications impose the shared representation

$$x_d = \bar{x} + \sum_{a=1}^{r} z_d^a u_a, \qquad r \in \{100, 200, 400\}, \tag{18}$$

and jointly estimate the mean $\bar{x}$, loadings u_a, and daily codes z_d. See Buehler et al. [2026a] for motivation and further results.

For model price C, quoted interval $[b, a]$, midpoint $m = (a+b)/2$, and half-spread $h = (a-b)/2$, define the bid–ask *excess*

$$e(C; b, a) = \frac{\max(C - a, 0) + \max(b - C, 0)}{h}.$$

Writing n_d for the number of quotes on date d, the quote component of the training objective is

$$\mathcal{L} = \frac{1}{891} \sum_{d} \frac{1}{n_d} \sum_{\ell=1}^{n_d} \left[e(C_{d\ell}; b_{d\ell}, a_{d\ell}) + 0.05 \frac{|C_{d\ell} - m_{d\ell}|}{h_{d\ell}} \right], \tag{19}$$

plus a soft barrier near the upper DLV-squared bound, cf. Buehler et al. [2026a]. Thus a price within its quoted interval has zero excess, while the small midpoint term identifies a stable point inside the spread. We use the exact all-quote gradient, accumulated in memory-bounded quote blocks, and Adam for up to 1,000 epochs.

Table 1 reports statistics pooled over all 2,192,458 retained quotes. Conditional excess is the mean of e only over quotes outside their spread; midpoint MAE is also measured in half-spreads.

Representation	State dim.	Inside bid–ask	Mean excess	Conditional excess	Midpoint MAE
Direct 25-by-200	5,000	97.32%	0.035	1.288	0.237
Affine-100	100	89.68%	0.083	0.808	0.471
Affine-200	200	95.02%	0.045	0.900	0.349
Affine-400	400	96.01%	0.040	1.009	0.317

Table 1: End-to-end fit to the retained market bid–ask intervals. State dimension is 5,000 for the unrestricted daily state and the latent rank for an affine state.

The direct representation gives the highest bid–ask coverage and the lowest midpoint error and is therefore used for the stochastic-DLV experiments below. Affine-200 already exceeds 95% bid–ask coverage; increasing the rank to 400 gives a smaller further improvement. Conditional excess is computed only on the changing set of failed quotes and should not be read as a monotone ranking across representations.

3.3 DLV vol-of-vol estimation

The floating historical supports cannot be differenced node by node without first aligning contracts. We use 710 adjacent market-session pairs ending no later than 31 December 2024. Write

$$L_d^j(x) := \log \Sigma_d^j(x), \qquad R_d := \frac{S_{d+1}}{S_d},$$

and let $x_{d,i}$ be the ith relative strike on the origin-date support. In the zero-carry normalization used for the dynamics, holding the corresponding cash strike fixed moves its relative strike to $x_{d,i}/R_d$ on the next date. Holding calendar expiry fixed simultaneously shifts the remaining-expiry index down by one. The aligned observation is therefore

$$\Delta L_d^{j,i} = L_{d+1}^{j}\left(\frac{x_{d,i}}{R_d}\right) - L_d^{j+1}(x_{d,i}), \qquad j = 1,\dots,24, \quad i = 1,\dots,200. \tag{20}$$

The endpoint value is interpolated on date $d+1$'s own floating support. Thus cash strike and calendar expiry are fixed within each one-day pair, while the origin chart is allowed to float across different pairs.

For each expiry–strike coordinate we fit the physical one-step regression

$$L_{d+1}^{j}\left(\frac{x_{d,i}}{R_d}\right) = a^{j,i} + b^{j,i} L_d^{j+1}(x_{d,i}) + \varepsilon_d^{j,i}. \tag{21}$$

The centered residual surfaces are smoothed by a Gaussian filter with widths one expiry node and two strike nodes. Let

$$Z_d = \Phi^{-1}\big(F_d^1(R_d)\big), \qquad \widetilde{Z}_d = \frac{Z_d - \overline{Z}}{s_Z},$$

where F_d^1 is the fitted front distribution including the independent 5% Black factor. The daily loading of the spot factor is

$$\ell_1^{j,i} = \frac{1}{n}\sum_d \widetilde{\varepsilon}_d^{j,i}\,\widetilde{Z}_d.$$

We remove this component and apply PCA to $\widetilde{\varepsilon}_d - \widetilde{Z}_d\ell_1$; its leading eigenvectors give the remaining daily loading surfaces $\ell_2, \ell_3, \dots$. The annualized DLV vol-of-vol used in the SDE is

$$\xi^{j,i;f} = \sqrt{252}\,\ell_f^{j,i}. \tag{22}$$

The fitted spot scores have mean 0.0290 and standard deviation 1.0026. Eight factors retain 72.88% of the smoothed innovation variance and 16 retain 84.40%; 26 factors are required to exceed 90%, reaching 90.25%. The rank-16 system is nested, with its first eight loading surfaces identical to the rank-8 system. Appendix A.2 shows the baseline implied-volatility surface, the cumulative factor power, and the implied- volatility response to each retained factor. On the held-out 2025 sample the full 25-by-200 representation has residual root mean square 0.256, compared with 0.423 for the Affine-200 representation.

This is a functional estimate in remaining-expiry and normalized-moneyness coordinates, not an estimate on one global historical cash-strike grid. When a simulation is initialized, these loading surfaces are attached nodewise to that state's fixed cash-strike grid. Pooling dates therefore assumes that the loading functions are stable in this moving coordinate chart.

The empirical physical log-DLV-squared drift implied by regression (21) is

$$\widehat{\mu}_{d,k}^{\mathbb{P},\mathrm{emp}} = 252\,[a_k + (b_k - 1)L_{d,k}], \tag{23}$$

where k denotes an aligned expiry–strike coordinate. At the same historical state we compute $\mu_d^{\mathbb{Q}}$ as the risk-neutral log drift in (13). Under the convention $\mathrm{d}W^{\mathbb{Q}} = \mathrm{d}W^{\mathbb{P}} + \lambda\,\mathrm{d}t$, a physical drift compatible with the fitted diffusion must have the form

$$\mu_d^{\mathbb{P},\mathrm{comp}} = \mu_d^{\mathbb{Q}} + \xi\lambda_d. \tag{24}$$

We estimate the state-dependent market price of risk by

$$\widehat{\lambda}_d = \arg\min_{\lambda \in \mathbb{R}^d} \sum_k w_{d,k} \left[\widehat{\mu}^{\mathbb{P},\mathrm{emp}}_{d,k} - \mu^{\mathbb{Q}}_{d,k} - (\xi\lambda)_k \right]^2 . \tag{25}$$

Here $w_{d,k}$ is proportional to the interior risk-neutral density, normalized so that each expiry has equal total weight. We also estimate one constant $\widehat{\bar{\lambda}}$ by pooling the same objective over all dates; this is the measure change used in the simulations. The independent 5% Black factor is assigned zero risk premium and therefore has the same law under $\mathbb{P}$ and $\mathbb{Q}$.

For comparison with a model-free time-series estimate, define the unconditional coordinate drift

$$\widehat{\bar{\mu}}^{\mathbb{P},\mathrm{unc}}_k = \frac{252}{n} \sum_{d=1}^{n} \Delta L_{d,k}. \tag{26}$$

Because (21) contains an intercept, this is also the sample mean of (23). We measure the distance of a compatible drift from this unconditional estimate by

$$D = \left[\frac{\sum_k \bar{w}_k \left(\overline{\mu^{\mathbb{P},\mathrm{comp}}}_k - \widehat{\bar{\mu}}^{\mathbb{P},\mathrm{unc}}_k \right)^2}{\sum_k \bar{w}_k} \right]^{1/2}, \qquad \bar{w}_k = \frac{1}{n} \sum_{d=1}^{n} w_{d,k}. \tag{27}$$

Calibration quantity	Rank 8	Rank 16
PCA innovation variance retained	72.88%	84.40%
$\mathbb{Q}$-drift RMS	13.58	13.77
Empirical $\mathbb{P}-\mathbb{Q}$ target RMS	39.41 (38.95, 39.88)	39.29 (38.82, 39.75)
Statewise compatible residual RMS	24.23 (24.01, 24.44)	20.81 (20.63, 20.99)
Statewise target energy captured	62.20% (61.64, 62.75)	71.95% (71.52, 72.37)
Constant-λ residual RMS	29.97 (29.58, 30.37)	28.54 (28.14, 28.94)
Constant-λ target energy captured	42.15% (40.94, 43.32)	47.24% (46.10, 48.37)
D to unconditional drift, statewise λ_d	16.33 (15.75, 16.90)	13.57 (13.14, 14.01)
D to unconditional drift, constant $\bar{\lambda}$	16.43 (15.80, 17.05)	13.73 (13.21, 14.26)

Table 2: Vol-of-vol coverage and physical-drift projection on the 710 one-session training pairs. Parentheses give 95% intervals. RMS quantities are annualized log-DLV-squared drifts. The target and residual intervals use 20,000 pair-bootstrap resamples; the intervals for D use the pair-level delta method. They condition on the fitted AR, PCA, and risk-neutral drift, so they quantify time-series sampling error rather than full re-estimation uncertainty.

The statewise fit is a diagnostic upper bound on what the retained diffusion span can reproduce. The constant-λ rows describe the more restrictive measure change actually used below. Rank 16 materially improves both fits, but neither retained factor system reproduces the empirical physical drift exactly. In particular, the confidence intervals in the last two rows do not include zero; the compatible drift remains measurably different from the raw unconditional estimate.

Appendix A.1 visualizes the effect of the drift change with four matched Model D paths (Section 4).

4 BHW No Dynamic Arbitrage Test

In this section we assess whether the model is indeed free of dynamic arbitrage. We are not aware of an accessible benchmark for dynamic arbitrage in the literature, so the methodology described here is a contribution in its own right, first presented in Buehler et al. [2026a].

We compare four model specifications:

Model	DLV factors	Steps/day
A	8	1
B	16	1
C	8	8
D	16	8

Every statistical audit uses 100,000 paths arranged as 50,000 independent antithetic pairs. The restricted-universe audits use a 20-business-day horizon; the full instrument-grid extension below uses 25 business days. For a live instrument i, the one-day return used in both diagnostics is

$$R^i_{p,n} = \frac{V^i_{p,n+1} - V^i_{p,n}}{Z_{p,n}}, \tag{28}$$

where Z is spot and V^i is either spot or an option price. Division by the pre-step spot is predictable and therefore preserves the zero-conditional-mean property under $\mathbb{Q}$.

4.1 Risk-Neutral Diagnostics

Numerically, it is clear that as the number of instruments increases on a fixed set of paths, the likelihood of a risk-free return of a linear combination of instruments simply due to sampling error increases.

We will therefore test absence of dynamic arbitrage on a reduced trading universe which consists of spot and 44 options with cash strikes fixed at the initial date: eleven OTM calls or puts at each of 5, 10, 15, and 20 business days. Thus 45 instruments are live initially, giving 20 spot and 550 option day–instrument coordinates over the 20-day horizon.

The resulting test will only confirm absence of dynamic arbitrage within this restricted set of instruments, and only within the class of strategies considered. To illustrate the limitations of the former, we apply the same test to an enlarged trading universe in Section 4.3.

Unconditional diagnostic. For every day–instrument coordinate (n, i), first average the two members of each antithetic pair and then form

$$z_{n,i} = \frac{\overline{R}_{n,i}}{\widehat{\mathrm{se}}(\overline{R}_{n,i})}. \tag{29}$$

The tables report the maximum and 99th percentile of $|z_{n,i}|$ and the number of coordinates beyond fixed thresholds. This checks the unconditional implication $\mathbb{E}^{\mathbb{Q}}[R^i_{n+1}] = 0$ separately at every day and instrument. It does not test $\mathbb{E}^{\mathbb{Q}}[R^i_{n+1} \mid \mathcal{F}_n] = 0$ at every simulated state. Under $\mathbb{P}$ the same zero-mean statistics describe fitted physical drift; they are not no-arbitrage violations. For one standard-normal coordinate the 99th percentile of $|z|$ is 2.576. Coordinate dependence and the simultaneous screen mean that the reported counts are diagnostics, not family-wise p-values. The 1.96 and 2.576 thresholds apply to each coordinate-level z-score, not directly to the maximum or empirical quantiles across all coordinates.

Conditional diagnostics. We split complete antithetic pairs into two equal path sets A and B, and write $\mathcal{I}_A$ for all live path–time observations from paths in A. The state $X_{p,n}$ contains the cumulative Brownian-factor histories, log spot, the live log option-price-to-spot ratios, calendar time, and time to each expiry. On set A we fit the uncapped affine-plus-deep action

$$a_\theta(X_n) = D_n \left[b + L\widetilde{X}_n + g_\theta(\widetilde{X}_n) \right], \tag{30}$$

where D_n masks expired instruments, $\widetilde{X}$ is standardized using set A only, and g_θ is a two-layer width-32 SiLU network.

For either diagnostic $\delta \in \{\mathrm{sqr}, \mathrm{ent}\}$, let $\widehat{a}^{\delta}_{-p}$ denote the action fitted on the fold not containing path p. Its out-of-fold score is

$$d^{\delta}_p = \frac{1}{H} \sum_{n=0}^{H-1} \widehat{a}^{\delta}_{-p}(X_{p,n})^\top R_{p,n}, \tag{31}$$

where $H = 20$ in the restricted experiment and $H = 25$ on the full instrument grid. The factor $1/H$ only fixes the unit: removing it multiplies both the estimate and its standard error by H. For every antithetic pair $r = (p, \bar{p})$ define $\widetilde{d}^{\delta}_r = (d^{\delta}_p + d^{\delta}_{\bar{p}})/2$, let $\overline{d}^{\delta}$ be the mean of the N_{pair} pair scores, and let s_δ be their sample standard deviation. We use the following two diagnostics, in this order.

(i) *Squared diagnostic.* On training fold A we maximize

$$\widehat{J}^{\mathrm{sqr}}_A(\theta) = \frac{1}{|\mathcal{I}_A|} \sum_{(p,n)\in\mathcal{I}_A} \left[a_\theta(X_{p,n})^\top R_{p,n} - \frac{1}{2}\|a_\theta(X_{p,n})\|^2 - \frac{\lambda}{2}\big(a_\theta(X_{p,n})^\top R_{p,n}\big)^2 \right], \qquad \lambda = 1. \tag{32}$$

Pointwise in the state, its unrestricted population optimizer is $(I+\lambda\mathbb{E}[RR^\top \mid X])^{-1}\mathbb{E}[R \mid X]$. Thus the realized squared portfolio return includes cross-instrument second moments without requiring a separately estimated covariance matrix. The squared action statistic is

$$T_{\mathrm{sqr}} := \frac{\overline{d}^{\mathrm{sqr}}}{s_{\mathrm{sqr}}/\sqrt{N_{\mathrm{pair}}}}. \tag{33}$$

(ii) *Entropy diagnostic.* On training fold A we minimize

$$\widehat{L}^{\mathrm{ent}}_A(\theta) = \log\left[\frac{1}{|\mathcal{I}_A|} \sum_{(p,n)\in\mathcal{I}_A} \exp\left(-a_\theta(X_{p,n})^\top R_{p,n} + \frac{1}{2}\|a_\theta(X_{p,n})\|^2 \right) \right]. \tag{34}$$

For conditionally Gaussian returns the population action is approximately[10]

$$a^\star_{\mathrm{ent}}(X_t) \approx (I + \mathrm{Cov}_t(R))^{-1}\mathbb{E}_t[R].$$

The entropy action statistic is

$$T_{\mathrm{ent}} := \frac{\overline{d}^{\mathrm{ent}}}{s_{\mathrm{ent}}/\sqrt{N_{\mathrm{pair}}}}. \tag{35}$$

[10] On the filtration generated by the tangent space of the optimal action.

In each case the action is fitted on A, frozen and evaluated on untouched B, after which A and B are exchanged and the two out-of-fold sets of pair scores are combined. Under the conditional zero mean null hypothesis every frozen predictable action has zero expected score, and both T_{sqr} and T_{ent} are approximately standard normal. Either statistic is shown in black when its absolute value is at most 1.96, orange when it lies in $(1.96, 2.576]$, and red when it exceeds 2.576. These are per-test confidence thresholds. Each is a conditional test relative to the state and strategy class in (30), rather than a proof over all measurable strategies.

For broad instrument universes, isolated large z-scores are expected under the null purely from multiple testing. We therefore report both the extremes and the number of coordinates beyond fixed thresholds, and compare the latter with the corresponding standard-normal frequency.

4.2 Restricted Universe

Table 3 shows the $\mathbb{Q}$ results for the restricted trading universe. Orange marks a moderate departure from the Gaussian benchmark and red would mark a five-standard-error violation in the coordinate-wise screen. The separate conditional-statistic rule is given above; all eight conditional entries are black.

Model	N	Max. $\lvert z\rvert$	99% $\lvert z\rvert$	$N_{2.576}$	N_5	T_{sqr}	T_{ent}
		Unconditional				Conditional	
A: rank 8, one step	570	2.716	2.464	3	0	1.394	1.401
B: rank 16, one step	570	3.663	3.618	10	0	1.037	1.040
C: rank 8, eight steps	570	2.793	2.373	4	0	1.755	1.753
D: rank 16, eight steps	570	2.996	2.967	22	0	0.039	0.040

Table 3: Restricted-universe, 100,000-path $\mathbb{Q}$ diagnostics. The first five numeric columns are coordinate-wise unconditional statistics; the final two are the two-fold out-of-fold conditional statistics. Here N_c is the number of coordinates with $\lvert z\rvert > c$.

No restricted coordinate exceeds five standard errors, and no conditional statistic exceeds 1.755 in absolute value. Model D has the richest factor representation, the finer time discretisation, and the conditional statistic closest to zero. Table 4 applies the same diagnostics to paths generated with the fitted physical drift. The entries are deliberately not colour coded: nonzero physical expected returns are intended evidence of the measure change, not arbitrage failures.

Model	N	Max. $\lvert z\rvert$	99% $\lvert z\rvert$	$N_{2.576}$	N_5	T_{ent}
		Unconditional				
A: rank 8, one step	570	62.138	59.780	570	570	90.038
B: rank 16, one step	570	342.470	328.811	570	570	646.645
C: rank 8, eight steps	570	61.002	58.398	570	570	89.205
D: rank 16, eight steps	570	336.819	325.038	570	570	624.560

Table 4: Restricted-universe, 100,000-path $\mathbb{P}$ diagnostics, reported against zero to display the predictable drift created by the fitted measure change.

All 570 coordinates exceed five standard errors under every physical-measure specification, and the out-of-fold entropy statistics range from 89.205 to 646.645. Thus the same diagnostics

that remain small under $\mathbb{Q}$ recover both unconditional and state-predictable drift after the fitted change to $\mathbb{P}$.

4.3 Full Instrument Grid

The preceding experiment deliberately trades few options. We now test Model C on spot and the complete fixed grid of 200 interior strikes at each of 25 expiries, for 5,001 possible instruments. Eligibility is predictable and path dependent: before each daily trade, a live option is included only when

$$-2 < \frac{\log(K/Z_{p,n})}{\widehat{\sigma}_{\mathrm{ATM}}\sqrt{T_j - t_n}} < 1, \qquad \widehat{\sigma}_{\mathrm{ATM}} = 9.52\%.$$

Thus short expiries contain fewer eligible strikes and the traded set changes with spot. Ineligible and expired actions are exactly zero. Each conditional policy maps 5,035 pre-trade state variables to the masked 5,001-dimensional action. Each path score in (31) is averaged over 25 rather than 20 daily transitions; this changes its units but not either conditional statistic.

Model	N_{inst}	N_{coord}	Max. $\lvert z\rvert$	99% $\lvert z\rvert$	$N_{2.576}$	N_5	T_{sqr}	T_{ent}
			Unconditional				Conditional	
C: rank 8, eight steps	5,001	62,569	4.140	2.583	635	0	0.843	0.843

Table 5: Full-instrument-grid, 100,000-path $\mathbb{Q}$ diagnostics. A daily coordinate enters the unconditional screen when it has at least 20 eligible path observations and positive Monte Carlo variance. Of 63,180 coordinates that are eligible on at least one path, 62,569 enter the screen.

At the two-sided 1% threshold, 625.69 exceedances are expected among 62,569 independent standard-normal statistics; the observed count is 635. The 188 coordinates above three standard errors are similarly close to the 168.9 expected count, and none exceeds five. Dependence across neighbouring strikes and expiries makes these benchmark counts descriptive rather than a formal joint test, but the unconditional screen shows no excess tail beyond ordinary multiple-testing variation.

The squared diagnostic gives a mean captured daily return of 3.64×10^{-8} spot units with pair-based standard error 4.32×10^{-8}, hence $T_{\mathrm{sqr}} = 0.843$. Its 95% confidence interval, $[-4.83 \times 10^{-8}, 1.21 \times 10^{-7}]$, contains zero. The entropy diagnostic gives the same rounded mean, standard error, confidence interval, and $T_{\mathrm{ent}} = 0.843$. The unconditional and both strategy-conditional diagnostics therefore remain consistent with risk neutrality on the full dynamically eligible instrument grid.

5 Conclusion

We have given an explicit continuous-time construction of a risk-neutral measure for the SANOS discrete local volatility representation of European option markets, extending the static SANOS surface of Buehler et al. [2026b] to a dynamically arbitrage-free model. The resulting SDLV process admits a closed-form risk-neutral drift for every discrete local volatility coordinate and an exact front-density/spot diffusion, in place of the numerical martingale measure used in Buehler et al. [2026a]. We further proved that the DLV dynamics admit a unique strong solution under the assumption of a square-integrable bound on the vol-of-vol (Theorem 2.1).

Estimated on 891 trading days of S&P 500 index option data, the model's static fit covers more than 95% of quoted bid/ask intervals, and the restricted-universe $\mathbb{Q}$ diagnostics of Section 4 confirm the theoretical martingale property up to time-discretisation error. On the full dynamically eligible instrument grid, both the coordinate-wise unconditional screen and the two out-of-fold conditional diagnostics remain consistent with the risk-neutral null. We further estimate the physical dynamics and the implied market price of risk, recovering a stable low-rank factor structure for the DLV vol-of-vol.

Natural next steps include applying the full-instrument diagnostic to the remaining factor ranks and time discretisations, studying the discrete-branch correction of Remark 2.4, and extending the estimation beyond short-dated S&P 500 index options to other asset classes and longer tenors.

Acknowledgements

We want to thank the team at Wharton for their WRDS `https://wrds-www.wharton.upenn.edu` portal and OptionMetrics `https://optionmetrics.com/united-states` for providing their IvyDB data via this service. Without this data our work would not have been possible.

Declaration of funding

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A Model and DLV factor visualizations

A.1 Consistent Drift

Figure 1 illustrates the fitted measure change using four matched Model D paths. Solid and dashed lines of the same colour use the same Gaussian draws under $\mathbb{Q}$ and $\mathbb{P}$. Besides spot, the illustration reports ATM one- and five-day implied volatility and the central log-strike skew

$$\mathrm{Skew}_\tau = \frac{\sigma_{\mathrm{imp}}(K = Z_t e^{h}, \tau) - \sigma_{\mathrm{imp}}(K = Z_t e^{-h}, \tau)}{2h}, \qquad h = 0.01. \tag{36}$$

The paths illustrate the fitted measure change; four paths are not a physical model validation.

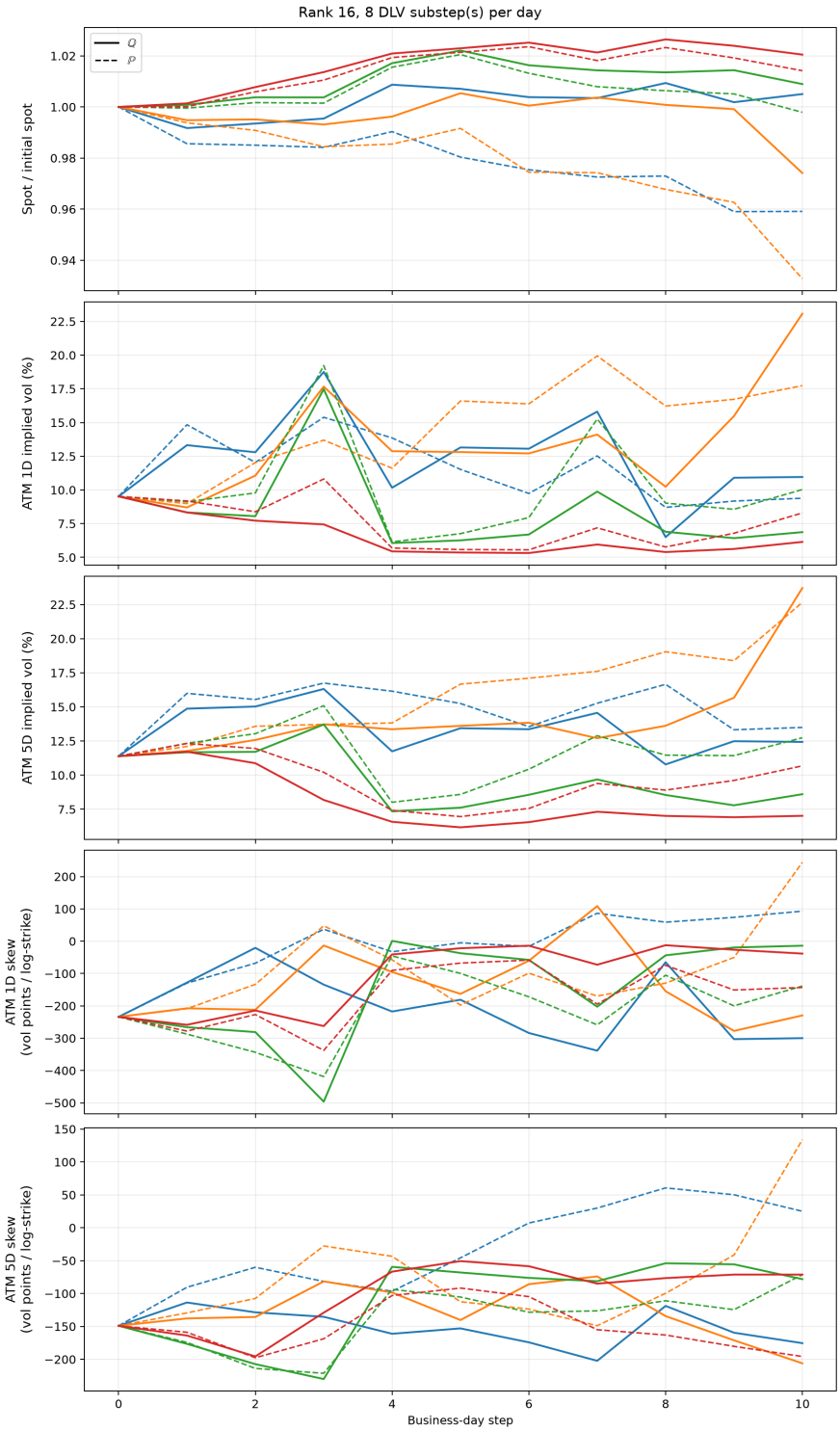


Figure 1: Matched $\mathbb{P}/\mathbb{Q}$ paths for Model D: rank 16 with eight Brownian substeps per day.

A.2 PCA Summary

The remaining figures visualize the factor systems estimated in Section 3.3. Figure 2 is the baseline risk-neutral implied-volatility surface used for all factor illustrations.

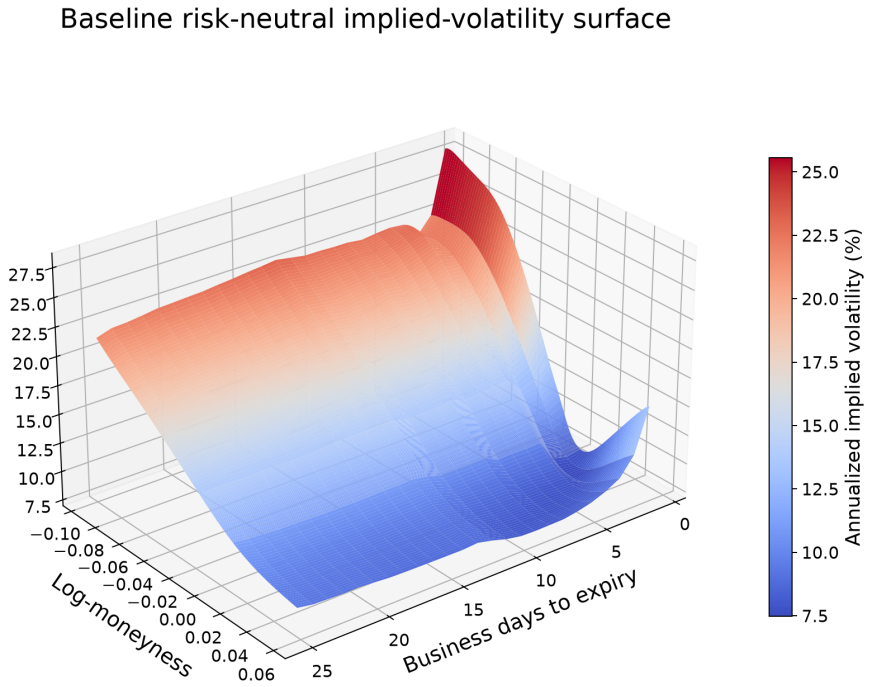


Figure 2: Baseline annualized implied volatility on the fitted log-moneyness and business-days-to-expiry grid.

Figure 3 reports individual and cumulative PCA power. The experiments retain either eight or 16 factors; the first rank above 90% is 26.

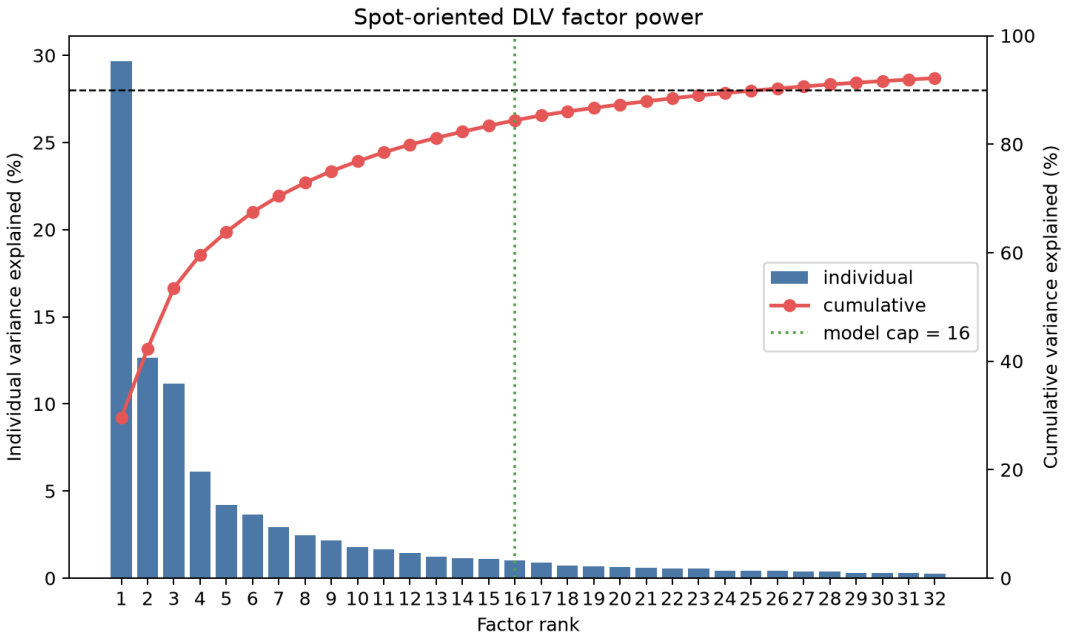


Figure 3: Individual and cumulative variance explained by the spot-oriented DLV factor system. The vertical line is the rank-16 model cap; 26 factors are required to exceed 90%.

The rank-16 fit appends PCs 9–16 to the rank-8 fit. A one-standard-deviation shock is the diffusion-only log-DLV displacement $\pm\xi_f\sqrt{1/252}$. The figures display its central implied-volatility response

$$\frac{1}{2}\left[\sigma_{\text{imp}}(\log\Sigma+\xi_f/\sqrt{252})-\sigma_{\text{imp}}(\log\Sigma-\xi_f/\sqrt{252})\right]. \tag{37}$$

These shock directions are the same under $\mathbb{P}$ and $\mathbb{Q}$: an equivalent measure change alters drift, not diffusion. Each panel deliberately retains its own vertical scale.

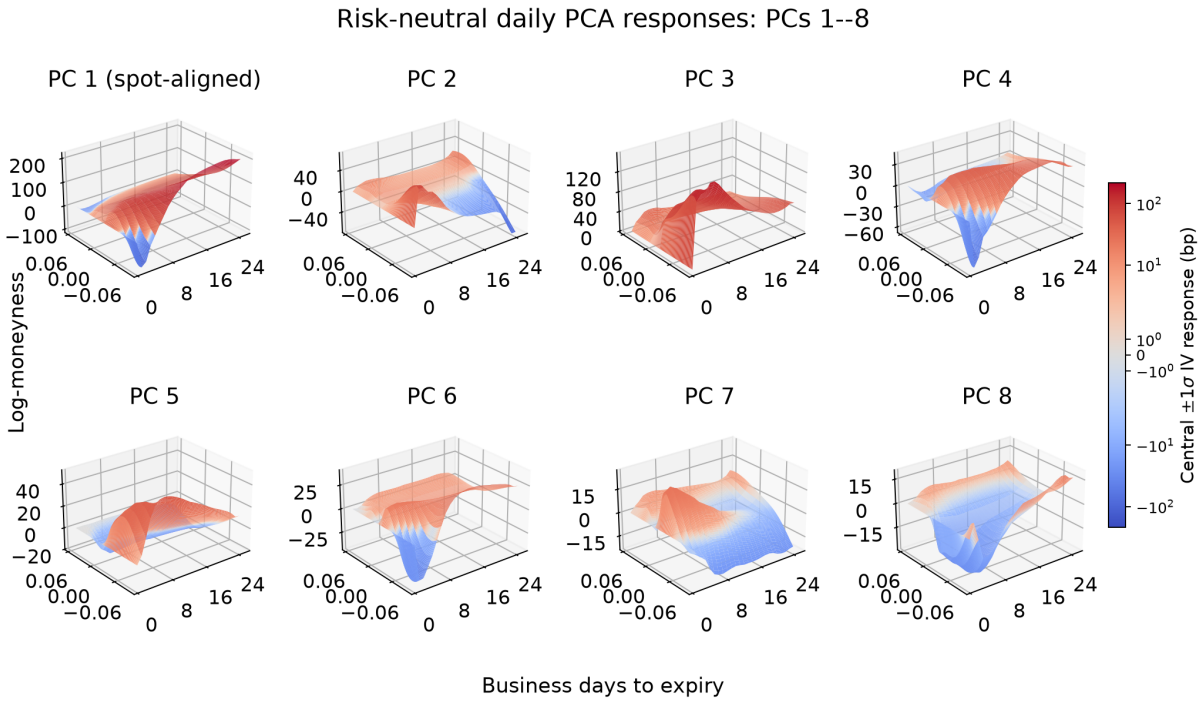


Figure 4: Risk-neutral one-standard-deviation implied-volatility responses for PCs 1–8. PC 1 is aligned to the front-spot score.

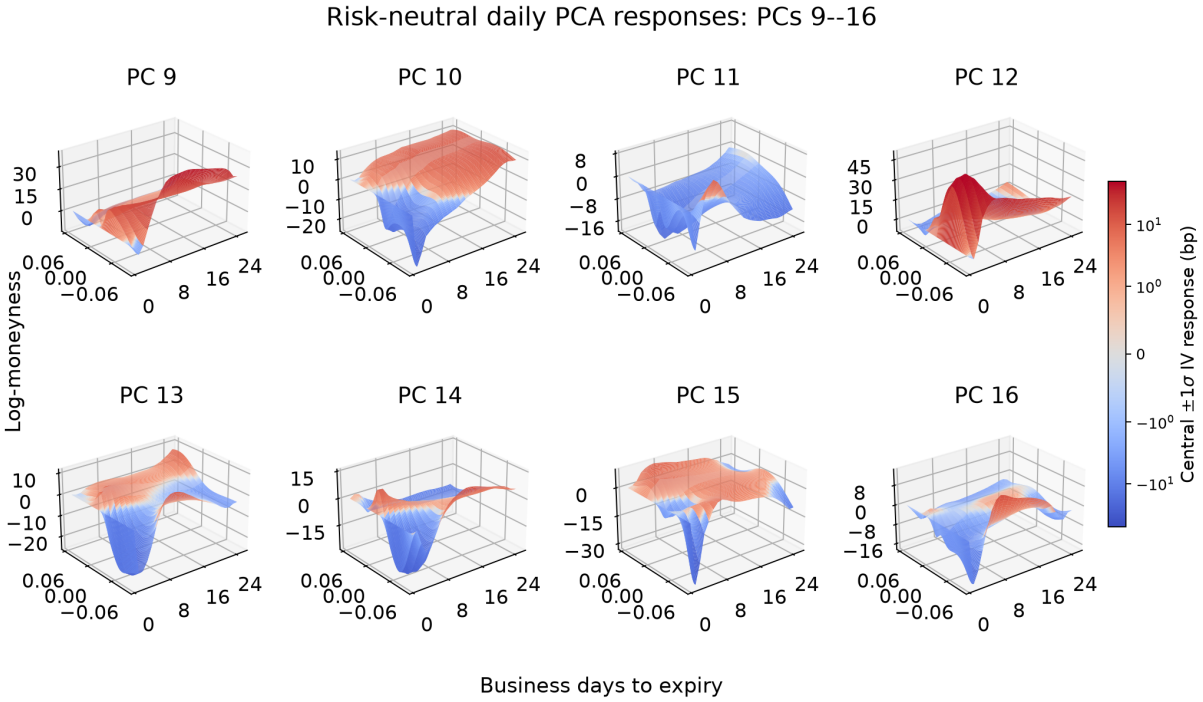


Figure 5: The additional one-standard-deviation implied-volatility responses, PCs 9–16, in the nested rank-16 model.

B Market-data and calibration pipeline

This appendix records the data and numerical pipeline behind Section 3. It follows the broader SPX pipeline described in Buehler et al. [2026a], restricted here to the short-dated fixed-expiry SANOS experiment. The complete transformation is

$$\begin{aligned}\text{raw IvyDB quotes} &\longrightarrow \text{daily cleaning and parity-forward enrichment}\\ &\longrightarrow \text{arbitrage-free SANOS eligibility surface}\\ &\longrightarrow \text{European short-horizon quote selection}\\ &\longrightarrow \text{daily 25-by-200 calibration tensor}\\ &\longrightarrow \text{direct or affine end-to-end fit.}\end{aligned}$$

B.1 Raw data and cleaning

The source is OptionMetrics IvyDB US data accessed through WRDS. For each eligible SPX market date we load the index level and previous index level, option strikes and calendar expiries, call and put bid and ask quotes, volume, open interest, AM/PM settlement flags, and the zero curve. The retained sample runs from 3 January 2022 through 29 August 2025 and contains 891 valid full-day NYSE sessions.

The raw cleaning step removes known bad ticker–date observations, missing or invalid quotes, invalid settlement flags, same-day expiries, irregular expiries, and quotes whose bid does not exceed 0.011. Business days and year fractions to expiry are then computed; an AM-settled expiry is treated as 0.2 of a business day earlier than its corresponding PM expiry.

For each expiry we recompute the forward from put–call parity rather than use the delayed IvyDB forward table. With discount factor D, define the discounted-forward bounds at a strike K by

$$G^{\mathrm{ask}}(K) = C^{\mathrm{ask}}(K) - P^{\mathrm{bid}}(K) + DK, \qquad G^{\mathrm{bid}}(K) = C^{\mathrm{bid}}(K) - P^{\mathrm{ask}}(K) + DK.$$

Their midpoint is averaged across strikes with inverse-spread weights,

$$\widehat{G} = \frac{\sum_K w_K \left[G^{\mathrm{ask}}(K) + G^{\mathrm{bid}}(K)\right]/2}{\sum_K w_K}, \qquad w_K = \frac{1}{\max\{G^{\mathrm{ask}}(K) - G^{\mathrm{bid}}(K), 10^{-3}\}}.$$

The resulting forward is used to construct the tightest call-equivalent quote, forward-normalized strikes and prices, implied volatilities, and Greeks. An expiry must contain at least ten strikes with a traded call or put and at least four strikes with both sides traded. The normalized spread must exceed 10^{-6} and retained implied volatilities must lie between 0.01 and 5.

B.2 Static preprocessing and quote selection

Before applying the model-specific horizon, the pipeline retains quotes with vega per unit total volatility of at least 10^{-3}, log volume and log open interest of at least one, and at least ten surviving options per expiry. The nominal per-day quote cap is 100,000 and is therefore non-binding here.

For data-quality control and to construct a stable ATM curve, every daily book is passed through the linear SANOS preprocessing of Buehler et al. [2026b]. The strike grid is extended, each expiry is fitted inside the quoted bid–ask intervals, negligible-density redundant strikes are removed, and the joint surface fit enforces both butterfly and calendar-spread constraints. The

smoothness parameter is 0.25. Expiries and strikes for which this arbitrage-free auxiliary fit is unavailable are not passed to the final calibration.

This auxiliary SANOS surface determines eligibility, the ATM term structure, and the floating model support. It is not the calibration target: the JAX fits in Section 3.2 use the original retained call and put bid–ask intervals. No linear-programming state or fitted midpoint is supplied as a target.

The final selection keeps European quotes no farther than 25 business days from the observation date, excludes the 0.2-business-day slice, and retains the standardized log-moneyness interval $[-2.5, 1.5]$. This leaves 2,192,458 quotes, between 1,590 and 5,096 per date.

B.3 Daily calibration tensor

The model expiry grid is

$$T_j = \frac{j}{252}, \qquad j = 1, \ldots, 25.$$

For each date, 100 interior-strike candidates are generated from the front ATM scale and 100 from the 25-day ATM scale. They are merged with a minimum log-spacing safeguard to form one 200-node support shared across all model expiries on that date. The support floats from date to date. Two boundary nodes, equal to one half of the lowest inner strike and one and a half times the highest inner strike, complete the 202-node density grid.

Each retained market quote stores its call/put flag, normalized strike, bid, ask, midpoint, business days to expiry, adjacent model-expiry indices, and linear interpolation weight. The contemporaneous SPX level is retained as a reference spot. The immutable calibration tensor pads each date to a multiple of 128 quotes and carries a Boolean mask, so padding never contributes to the objective.

At valuation time the DLV recursion produces the 25 discrete densities. The fixed 5% Black factor convolves each discrete density into a smooth call-price curve, and the two adjacent expiry curves are linearly interpolated to the market quote's exact expiry. Puts follow from put–call parity.

B.4 Optimization and retained artifacts

The direct and affine parameterizations are those in Section 3.2. Direct states are initialized from the date-specific ATM term structure and are constant across strike at initialization. Affine means use the average of these initial states; codes and loadings receive small centered Gaussian initial values. DLV squared is bounded between 10^{-4} and 9, with a differentiable barrier near the upper bound.

The loss in (19) is differentiated over every retained quote. Quote blocks of size 128 are used only to limit accelerator memory: their gradients are summed before each Adam update, so this is an exact full-sample gradient rather than stochastic minibatching. Each fit uses 300 initial epochs with a cosine learning-rate schedule from 2×10^{-2} to 10^{-3}, followed by a 700-epoch continuation from 5×10^{-3} to 5×10^{-5}. Checkpoints store either all daily direct states or the affine mean, loadings, and daily codes. The retained reports recompute every quote from the final checkpoint and produce the statistics in Table 1.

C Option-surface Literature Classification

Tables 6, 7, 8, 9, and 10 separate four questions which are often conflated: constructing an option or implied volatility surface, compressing or generating a market state, solving an underlying-model inverse problem, and learning a surface's evolution through time. We use common classification symbols throughout the tables:

- **S∞**: a theorem- or architecture-level guarantee on the full continuous surface.
- **SΩ**: the same guarantee only on a stated bounded fitting domain.
- **SG**: a strict guarantee only on the modelled finite grid.
- **A**: approximate, using penalties, reweighting, rejection, or empirical violation rates.
- **N**: no static-arbitrage guarantee.
- **U**: arbitrage freedom is claimed, but the available construction does not suffice to verify its continuum scope.

Table 6: Selected parametric implied-volatility and option-price interpolation methods. "Parametric" means that the surface is restricted to a prescribed finite-dimensional functional family; it does not describe the calibration algorithm. Constraints imposed during fitting are distinguished from post-fit repair.

Method	Representation	Static arb.	Native inputs	Off-grid	Scope and qualification
Gatheral [2004]	SVI smile	N by itself	Arbitrary, one expiry	∞ per expiry	Introduces raw SVI and its wing asymptotics, but does not supply the later cross-expiry SSVI construction or a complete admissible parameter domain.
Hafner [2004]	Polynomial IV surface	N	Variable daily DAX transactions	∞ on bounded domain	Weighted least-squares fit followed by finite-grid screening: dates violating hedge, bull-spread, butterfly, or calendar tests on a 200-point grid are discarded. The fitted family itself is unconstrained.
Bloch [2013]	Mixture family	U	Arbitrary	∞	Claimed arbitrage-free by construction; the continuum conditions were not verified.
Gatheral and Jacquier [2014]	SVI/SSVI	S∞	Arbitrary	∞	Strict for admissible parameters; a central low-dimensional benchmark.
Guo et al. [2016]	Generalized SVI	S∞	Arbitrary	∞	Extends the strict SVI/SSVI constructions.
Corbetta et al. [2019]	SSVI slices	S∞	Arbitrary maturities and strikes	∞	Calibrates butterfly- and calendar-consistent slices, then proves that its parameter interpolation and extrapolation preserve those conditions over the continuous surface.
Hendriks and Martini [2019]	eSSVI	S∞ (conditional)	Arbitrary	∞	Explicit butterfly and calendar conditions; strictness requires admissible parameter curves.
Martini and Mingone [2022]	SVI smile	S∞ per expiry	Arbitrary, one expiry	∞ per expiry	Characterizes the no-butterfly domain of raw SVI and calibrates directly on it; calendar consistency across independently fitted expiries is not automatic.
Mingone [2022]	eSSVI surface	S∞	Arbitrary	∞	A global eSSVI parametrization whose admissible parameter domain enforces butterfly and calendar-spread conditions.
Roos [2025]	Discrete-time local-vol family	U	Arbitrary coterminal set	∞	The abstract claims analytic strike interpolation and expiry extension; the continuum inequalities were not verified.

Table 7: Selected non-parametric implied-volatility and option-price interpolation methods. Soft-penalty methods are approximate even when reported violations are negligible.

Method	Representation	Static arb.	Native inputs	Off-grid	Scope and qualification
Kahalé [2004]	Piecewise analytic call/IV interpolation	SΩ (conditional)	Arbitrage-free discrete quotes	∞ for $T \leq T_{\max}$	Convex-envelope construction on the bounded maturity domain; no extension beyond the last input expiry.
Fengler [2009]	Constrained spline	SΩ per expiry; SG calendar	Arbitrary	∞ on fitted strike domain	Strike constraints hold over each fitted spline; calendar constraints only at knots and input maturities.
Andreasen and Huge [2011]	One-step implicit FD; basis for DLV	SG numerically; S∞ for exact local-vol model	Arbitrary quotes	∞	Sequential local-vol calibration; the finite-difference scheme preserves monotonicity and convexity on its grid.
Orosi [2015]	Thin-plate regression splines/QP	U	Arbitrary	∞	Quadratic-program constraints; their continuum scope was not verified.
Ackerer et al. [2020]	Neural network	A	Irregular daily panel	∞	Static conditions enter through soft penalties; violation rates depend on the penalty weight.
Bender and Thiel [2020]	Reference-model distortion of call prices	S∞ (conditional)	Arbitrage-free discrete quotes	∞	Smooth strike interpolation and arbitrage-free maturity interpolation.
Zhang et al. [2023]	PCA/VAE/samples plus neural surface	A	$14 \times 11 = 154$	∞	Calendar, butterfly, and wing conditions enter as penalties; occasional spikes remain.
Chataigner et al. [2021]	Constrained GP or penalized NN	GP: SΩ; NN: A	Arbitrary	∞	Nodal inequalities imply monotonicity and convexity throughout the GP's bounded piecewise-multilinear domain. The NN and fitted SSVI alternatives use penalties.
Deschâtres [2024]	Variance-space cubic/B-spline QP	A	Arbitrary; 14,500 quotes in the SPX example	∞	Calendar constraints are linear, but butterfly constraints are linearized and the QP is iterated. Under the present convention this is approximate rather than a single exact admissible fit.
Yang et al. [2025]	Hypernetwork plus compact neural surface	A	9 observations	∞	Reconstructs index-option IV surfaces from sparse option quotes. Calendar and density conditions are finite-grid auxiliary losses; the paper explicitly states that they provide no hard guarantee.
Wiedemann et al. [2025]	Graph neural operator	A	Irregular intraday panels	∞ on its output domain	Calendar and butterfly conditions are finite-difference penalty losses evaluated at sampled coordinates; the paper describes the output as virtually arbitrage-free, not structurally invariant.
Schlenkrich [2025]	Piecewise shifted-Black/Bachelier model	S∞ per expiry; calendar N	Arbitrary	∞ per expiry	Whole-strike construction for one terminal distribution; calendar consistency is not addressed.
Qin et al. [2026]	Piecewise-quadratic calls/RN marginals	S∞ in strike at quoted expiries	Arbitrage-free discrete quotes	∞ in strike only	Explicit sequential completion of risk-neutral marginals; no interpolation between expiries.
Buehler et al. [2026b]	Smooth non-parametric DLV	S∞	Arbitrary; about 1,000 in example	∞	Bid–ask-aware smooth surface with positivity as the essential parameter constraint.

Table 8: PCA, autoencoder, and other latent representations of option surfaces. "Off-grid" counts options not represented by native model coordinates; external interpolation is not credited to the representation. An off-grid count of ∞ means that the representation can be evaluated on a continuum, subject to its stated domain and tail assumptions.

Method	Representation	Native option coordinates	Off-grid	Latent/static arb.	Qualification
Skiadopoulos et al. [2000]	Direct-IV PCA	Not reported; three maturity groups	0	N	PCA factors do not preserve static inequalities.
Cont et al. [2002]	Functional/direct-IV PCA	Not reported	Any number within the fitted domain	N	Functional interpolation does not make the PCA cone invariant.
Fengler et al. [2003]	Common PCA of direct IV changes	$6 \times 6 = 36$: moneyness 0.85–1.10 at 1, 2, 3, 6, 9, and 12 months (DAX)	0	N	Common factors summarize smile dynamics but do not define an invariant no-arbitrage state space.
Hafner [2004]	Four parametric IV factors: level, slope, curvature, and term slope	Variable daily DAX transactions; 200-point diagnostic grid	∞ on its bounded domain	N	The daily regression family is not constrained. Dates failing finite-grid hedge, bull-spread, butterfly, or calendar tests are discarded ex post.
Panigirtzoglou and Skiadopoulos [2004]	Separate implied-distribution PCA by expiry	$3 \times 21 = 63$: 21 deltas at 1, 3, and 6 months	0	PDF: S∞ per expiry; CDF: N; joint surface: N	The PDF construction is strict per expiry; neither construction enforces calendar consistency.
Dellaportas and Mijatović [2014]	Parametric SABR encoding	SABR triplet for one expiry	Any strike in fitted interval	U	SABR may violate arbitrage conditions in the extreme wings.
Wiese et al. [2019]	DLV, PCA, and GAN	$8 \times 4 = 32$	0	SG	Finite DLV lattice.
Avellaneda et al. [2020]	PCA of normalized IV-surface deformations	$8 \times 7 = 56$ per name: 8 maturities and 7 deltas	0	N	A data-driven factor representation; no static-arbitrage invariant decoder is imposed.
Bergeron et al. [2021]	VAE representation of IV surfaces	$8 \times 5 = 40$: 8 maturities and 5 deltas (FX)	Any (delta, maturity) in the domain via the pointwise decoder	N	A hands-off latent representation and generator, without a hard static-arbitrage architecture.
Wiese et al. [2021]	DLV, constrained autoencoder, and flow	$9 \times 4 = 36$ per asset	0	SG	Decoder is constrained on the finite lattice.
Zhang et al. [2023]	PCA/VAE/samples plus neural reconstruction	154	∞	A	Reconstruction uses static-arbitrage penalties.
Andrès et al. [2026]	Four-parameter SSVI state	384 (Euro Stoxx 50); 1,944 (SPX, before filtering)	∞	S∞	Admissible SSVI parameter conditions hold for each state.
Cohen et al. [2023]	Affine factors of normalized call prices	46 options on an irregular rolling (τ, m) lattice	0	SG	Linear no-static-arbitrage inequalities define a factor polytope.
Cont and Vuletić [2023]	Direct-IV PCA/factors plus path reweighting	Fixed IV grid	0	Raw PCA: N; output: A	Arbitrage scenarios are downweighted, not excluded structurally.
Choudhary et al. [2024]	Functional PCA and neural SDE	$17 \times 11 = 187$ per asset	Any number within the fitted domain	N	Static-arbitrage violations are measured only ex post.
Vuletić and Cont [2024]	Direct-IV GAN plus penalties/reweighting	$10 \times 8 = 80$	0	A	Approximate output correction.
Jin and Agarwal [2025]	Direct-IV conditional diffusion	$9 \times 9 = 81$	0	A	Penalty-based.
Golubitsky et al. [2026]	VAE latent representation	$2 \times 11 \times 17 = 374$	0	A	Static constraints are approximate.
Brooks et al. [2026]	Arbitrage-penalized VAE and latent flow	$32 \times 16 = 512$	0	A; 90.8% pass all tests	Empirical pass rate, not a guarantee.
Hao and Ji [2026]	Conditional diffusion plus SAAM refinement	$15 \times 17 = 255$	Arbitrary target coordinates	A	SAAM uses residual regularization and reduces measured violations on a finite diagnostic grid; it is not an admissible parameterization.
Han et al. [2026]	Joint return/direct-IV adaptive sequential diffusion	$11 \times 9 = 99$ IVs	0	A	Fine-tuning penalizes calendar-, call-, and butterfly-spread violations; reported violations are nearly zero, not structurally excluded.
Yang et al. [2026]	Joint stock-return/option-IV diffusion across equities	$11 \times 9 = 99$ option IVs per stock	0	A	Transfers dynamics to unseen stocks on the same fixed grid.
This paper	Smooth SANOS plus affine latent state	$18 \times 200 = 3{,}600$ prices; 3,620 controls	∞	S∞	The decoded continuum remains in the SANOS admissible family.

Table 9: Machine-learning methods for underlying-model inverse problems. These methods infer or optimize parameters of a pre-specified pricing model; they do not make the observed option market itself the dynamic state. Static arbitrage properties are inherited from the pricing model, subject to decoded parameters remaining admissible.

Method	Underlying pricing family	Role of ML	Online inverse step	Why separate from a market-state model
Hernandez [2016]	Hull–White/G2++ examples	Direct inverse map trained on model-generated examples	One network evaluation	Maps quotes to parameters of a prescribed model.
Dimitroff et al. [2018]	Heston	Convolutional network maps volatility grids to model parameters	One network evaluation	Amortized inversion into a prescribed pricing family.
Bayer and Stemper [2018]	Rough Bergomi and related rough-volatility models	Learns the forward parameter-to-IV map	Classical Levenberg–Marquardt optimization through the surrogate	The NN accelerates a conventional model calibration.
Horvath et al. [2019], Bayer et al. [2019]	Broad class including rough-volatility models	Learns a fast forward pricing/IV map	Classical numerical or Bayesian inversion of the surrogate	Explicitly retains the two-step pricing-model-plus-calibration architecture.
Liu et al. [2019]	Heston and Bates	Learns a forward parameter-to-observable map	Differential evolution through the surrogate	The learned map accelerates conventional parameter calibration.
Itkin [2019]	Black–Scholes examples	Compares direct and surrogate calibration	Direct map or numerical solver	Studies smooth Greeks, soft arbitrage penalties, and inverse-map stability.
Roeder and Dimitroff [2020]	Volatility-model examples used in deep-calibration benchmarks	Compares a direct IV-to-parameter map with the two-step forward surrogate	None for the direct version; solver for the indirect version	Both outputs are parameters of an exogenously specified pricing model.
Cuchiero et al. [2020]	Local stochastic-volatility models	Neural leverage function	Monte Carlo SGD; adversarial optimization only for robust bid–ask calibration	The generator remains an underlying-first LSV model.
Rosenbaum and Zhang [2021]	Quadratic rough Heston	Neural pricing/calibration engine based on a multifactor approximation	Optimization against the learned engine	The calibrated state is a rough-volatility parameter vector.
Ning et al. [2023]	Time-inhomogeneous CTMC; additive Gaussian-mixture and CGMY/KoBoL Lévy models	VAE/PCA learns the cross-date distribution of already calibrated parameters	Regularized inverse calibration is done first, date by date; CTMC maturities are fitted sequentially	The VAE generates model parameters, and prices are recovered through the model's characteristic function.
Sridi and Bilokon [2023]	Heston	Neural pricing surrogate	Differential evolution and Nelder–Mead	Heston supplies the risk-neutral structure.
Zhang et al. [2024]	Heston	Differential network learns prices and sensitivities	Gradient calibration through the learned pricer	Heston supplies the risk-neutral structure.

Table 10: Option-surface dynamics under the historical measure ¶ and direct market-model dynamics under $\mathbb{Q}$. The $\mathbb{Q}$ column excludes underlying-first pricing families such as Heston, local volatility, and Bates. "Multi-step ¶ path" records whether the cited work actually reports a simulated or forecast path spanning more than one transition, rather than whether its model could in principle be iterated. "Implemented" refers to the stated dynamic construction, not merely static preprocessing.

Method	Dynamic state/model	Static arb. under ¶	Multi-step ¶ path	Direct market-model dynamics under $\mathbb{Q}$	Implemented	Qualification
Schönbucher [1999]	One fixed-strike term structure of implied volatilities	—	—	Exact drift restrictions across maturities for one strike	MC implementation strategy only	The setup initially permits a generic finite option set, but the developed discrete and continuous constructions vary maturity at one strike. Independent dynamics across strikes at the same maturity are explicitly left for future research; this is not a full-surface model.
Skiadopoulos et al. [2000]	Direct-IV PCA factors	N	No	—	Yes	No static invariant state space.
Cont et al. [2002]	Functional/direct-IV PCA factors	N	No	— (¶ only)	Empirical PCA only	No constructed admissible $\mathbb{Q}$ dynamics.
Cont and da Fonseca [2002]	Functional PCA factors of direct IV surfaces	N	No	—	Yes (¶)	Empirical factor dynamics; distinct from the three-author stochastic-surface paper and not a constructed risk-neutral model.
Fengler et al. [2003]	Common PCA factors of direct IV changes	N	No	—	Yes (¶)	Empirical CPCA of observed daily surfaces. The factors are proposed as a starting point for VaR Monte Carlo, but the paper does not report synthetic multi-step IV paths. There is no invariant static-arbitrage state space.
Hafner [2004]	Four IV factors with correlated OU dynamics	N; screened sample only	No (1-, 5-, and 10-day forecasts)	A/local	Yes	The fitted OU model violates the global drift condition; the $\mathbb{Q}$ construction is local and approximate.
Panigirtzoglou and Skiadopoulos [2004]	Separate distribution factors by constant maturity	PDF: S∞ per expiry; CDF: N	Yes (14 days)	N for a joint surface	Yes	Neither construction enforces calendar consistency.
Dellaportas and Mijatović [2014]	Single-expiry SABR parameter process	U	No (1–3-day point forecasts)	—	Yes (¶)	No joint surface; extreme-wing arbitrage is not excluded.
Fengler et al. [2007]	Semiparametric IV-factor surface	N	No	—	Yes (¶)	Estimates and forecasts historical surface factors without a hard static-arbitrage invariant.
Schweizer and Wissel [2008a]	Price level and local implied-volatility field	S∞, theoretical	—	S∞: solvability theorem and explicit admissible coefficient examples	No	One fixed expiry and all strikes; not a finite option lattice or full maturity surface.
Schweizer and Wissel [2008b]	Term structure for one fixed payoff	S∞, theoretical	No	Strict theoretical drift/existence results	No	Existence for fixed-strike calls and a power payoff; not a full-surface model.
Carmona and Nadtochiy [2009]	Local-volatility surface as option-market codebook	—	—	Drift and spot-consistency characterization	Static empirical fit only	Direct HJM-style codebook dynamics; general model existence is not proved.
Carmona and Nadtochiy [2012]	Tangent Lévy option-market codebook	—	—	S∞ under the consistency conditions	Theory	Constructs direct risk-neutral dynamics for all strikes and maturities from admissible tangent Lévy characteristics.
Kallsen and Krühner [2015]	Time-inhomogeneous Lévy option codebook	—	—	S∞: necessary/sufficient conditions and existence	No numerical implementation	Constructive existence and uniqueness from stated building blocks, but a theoretical HJM framework rather than a calibrated simulator.

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Table 10 – continued

Method	Dynamic state/model	Static arb. under ¶	Multi-step ¶ path	Direct market-model dynamics under $\mathbb{Q}$	Implemented	Qualification
Carmona et al. [2017]	Finite-dimensional tangent Lévy specifications	—	No	S∞ under model admissibility	Yes	Risk-neutral HJM drift with PCA-estimated volatility; eight-day $\mathbb{Q}$ paths.

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Table 10 – continued

Method	Dynamic state/model	Static arb. under ¶	Multi-step ¶ path	Direct market-model dynamics under $\mathbb{Q}$	Implemented	Qualification
Wiese et al. [2019]	DLV PCA/GAN state	SG	Yes	—	Yes	Finite-grid historical dynamics.
Cao et al. [2020]	Neural regression of one-day IV changes	N	No	—	Yes (¶)	Expected changes depend on return, delta/moneyness, maturity, and VIX; no arbitrage constraint.
Wiese et al. [2021]	Constrained DLV latent flow	SG	Yes	—	Yes	Finite-grid historical dynamics.
Zhang et al. [2023]	LSTM latent evolution and neural reconstruction	A	No (next-day only)	—	Yes	Static-arbitrage penalties only.
Buehler et al. [2022]	General ¶-simulator plus an OCE/entropy path-density model	Inherited from the input simulator	Input-dependent	Numerical near-martingale measure for a finite trading universe	Yes	Reweights an input simulator; with frictions, drift is targeted within transaction costs for selected instruments.
Medvedev and Wang [2022]	Direct-IV LSTM/ConvLSTM	N	No (multihorizon forecasts)	—	Yes (¶)	No static-arbitrage constraint.
Andrès et al. [2026]	Path-dependent SSVI factors	S∞ for every simulated snapshot	Yes	No market-measure dynamics	Yes (¶)	Static arbitrage freedom does not imply dynamic no-arbitrage.
Cohen et al. [2023]	Rolling normalized-call lattice, affine factors, and neural SDE	SG under ¶: exact for model SDE	Yes	— (not implemented)	Yes (¶ only)	The drift points inward and the normal diffusion vanishes at every face, so the continuous-time factor process remains in the finite-lattice no-static-arbitrage polytope ¶-a.s. The coordinates float with fixed (τ, m) rather than tracking fixed contracts. The HJM expression is only a factor-selection criterion and residual diagnostic, not an implemented $\mathbb{Q}$ model.
Cont and Vuletić [2023]	Direct-IV factors plus path reweighting	A	Yes	—	Yes	Reweighting reduces violations.
Choudhary et al. [2024]	Functional PCA and neural SDE	N	Yes	—	Yes	Violations are measured only ex post.
Vuletić and Cont [2024]	Direct-IV GAN dynamics	A	No (next-day only)	—	Yes	Penalties and reweighting.
Jin and Agarwal [2025]	Conditional diffusion on IV grids	A	No (next-day only)	—	Yes	Penalty-based.
Brooks et al. [2026]	Arbitrage-penalized VAE and latent flow matching	A	No	—	Yes	Only 90.8% of generated surfaces pass all reported tests.
Hao and Ji [2026]	Conditional diffusion plus SAAM refinement	A	Yes (five one-minute steps)	—	Yes	The daily experiment is next-day only; finite-grid violations remain.
Han et al. [2026]	Joint daily return and direct-IV grid via adaptive sequential diffusion	A	Yes (reported three-day path)	—	Yes (¶)	Each step conditions on the updated generated history; hedging uses one-day conditional scenarios.
Yang et al. [2026]	Joint stock-return/option-IV diffusion across equities	A	No (next-day only)	—	Yes	Jointly generates the next-day stock return and 11×9 option-IV increment; static option constraints are approximate.
François et al. [2026]	Five-factor IV surface jointly with SPX returns	A	No (five-day terminal scenarios)	—	Yes (¶)	Rare empirical violations, not an invariant state space.
Buchegger and Gonon [2026]	Latent diffusion for returns and IV surfaces	A	Yes (30-day trajectories)	—	Yes (¶)	Penalty-based arbitrage awareness; violations remain.
This paper	SANOS, affine latent state, and AR(1)	S∞	Yes	Limited SG: 48 options plus spot	Yes	The implemented $\mathbb{Q}$ construction is finite-dimensional.

References

Damien Ackerer, Natasa Tagasovska, and Thibault Vatter. Deep smoothing of the implied volatility surface. In *Advances in Neural Information Processing Systems*, volume 33, pages 11552–11563, 2020.

Jesper Andreasen and Brian Huge. Volatility interpolation. *Risk*, 24(3):76–79, 2011.

Hervé Andrès, Alexandre Boumezoued, and Benjamin Jourdain. The implied volatility surface (also) is path-dependent. *Quant. Finance*, 26(7):1017–1047, 2026. doi: 10.1080/14697688.2026.2637739.

Marco Avellaneda, Brian F. Healy, Andrew Papanicolaou, and George Papanicolaou. PCA for implied volatility surfaces. *J. Financial Data Sci.*, 2(2):85–109, 2020. doi: 10.3905/jfds.2020.1.032.

Christian Bayer and Benjamin Stemper. Deep calibration of rough stochastic volatility models, 2018. URL `https://arxiv.org/abs/1810.03399`.

Christian Bayer, Blanka Horvath, Aitor Muguruza, Benjamin Stemper, and Mehdi Tomas. On deep calibration of (rough) stochastic volatility models, 2019. URL `https://arxiv.org/abs/1908.08806`.

Christian Bender and Matthias Thiel. Arbitrage-free interpolation of call option prices. *Stat. Risk Model.*, 37(1–2):55–78, 2020. doi: 10.1515/strm-2018-0026.

Maxime Bergeron, Nicholas Fung, John Hull, and Zissis Poulos. Variational autoencoders: A hands-off approach to volatility, 2021.

Daniel Alexandre Bloch. Arbitrage-free mix var volatility surfaces and applications. SSRN working paper 2197185, 2013. URL `https://papers.ssrn.com/abstract=2197185`.

Alan Brace, Dariusz Gatarek, and Marek Musiela. The market model of interest rate dynamics. *Mathematical Finance*, 7(2):127–155, 1997. doi: 10.1111/1467-9965.00028.

Oscar Brooks, Dusica Bajalica, Yating Liu, and Imen Ben Tahar. Latent flow matching for arbitrage-aware implied volatility surface generation, 2026.

Dominik Manuel Buchegger and Lukas Gonon. Arbitrage-aware multi-step forecasting of implied volatility surfaces: Modelling surface trajectories using latent diffusion, 2026. URL `https://arxiv.org/abs/2608.22478`.

Hans Buehler and Evgeny Ryskin. Discrete local volatility for large time steps (extended version). SSRN working paper, 2015. URL `https://papers.ssrn.com/abstract=2642630`.

Hans Buehler, Phillip Murray, Mikko S. Pakkanen, and Ben Wood. Deep hedging: Learning to remove the drift. *Risk*, February 2022. URL `https://www.risk.net/cutting-edge/banking/7932226/deep-hedging-learning-to-remove-the-drift`.

Hans Buehler, Blanka Horvath, and Anastasis Kratsios. DYSANOS: Generative dynamic smooth arbitrage-free non-parametric option surfaces, 2026a. URL `https://arxiv.org/abs/2608.12587`.

Hans Buehler, Blanka Horvath, Anastasis Kratsios, Yannick Limmer, and Raeid Saqur. SANOS: Smooth strictly arbitrage-free non-parametric option surfaces, January 2026b. URL https://arxiv.org/abs/2601.11209. submitted.

Jay Cao, Jacky Chen, and John Hull. A neural network approach to understanding implied volatility movements. *Quant. Finance*, 20(9):1405–1413, 2020. doi: 10.1080/14697688.2020.1750679.

René Carmona and Sergey Nadtochiy. Local volatility dynamic models. *Finance Stoch.*, 13(1):1–48, 2009. doi: 10.1007/s00780-008-0078-4.

René Carmona and Sergey Nadtochiy. Tangent lévy market models. *Finance Stoch.*, 16:63–104, 2012. doi: 10.1007/s00780-011-0158-8.

René Carmona, Yi Ma, and Sergey Nadtochiy. Simulation of implied volatility surfaces via tangent Lévy models. *SIAM J. Financial Math.*, 8(1):171–213, 2017. doi: 10.1137/15M1015510.

Marc Chataigner, Areski Cousin, Stéphane Crépey, Matthew Dixon, and Djibril Gueye. Beyond surrogate modeling: Learning the local volatility via shape constraints. *SIAM J. Financial Math.*, 12(3):SC58–SC69, 2021. doi: 10.1137/20M1381538.

Vedant Choudhary, Sebastian Jaimungal, and Maxime Bergeron. FuNVol: A multi-asset implied volatility market simulator using functional principal components and neural SDEs. *Quant. Finance*, 24(8):1077–1103, 2024. doi: 10.1080/14697688.2024.2396977.

Samuel N. Cohen, Christoph Reisinger, and Sheng Wang. Arbitrage-free neural-SDE market models. *Appl. Math. Finance*, 30(1):1–46, 2023. doi: 10.1080/1350486X.2023.2257217.

Rama Cont and José da Fonseca. Dynamics of implied volatility surfaces. *Quant. Finance*, 2(1):45–60, 2002. doi: 10.1088/1469-7688/2/1/304.

Rama Cont and Milena Vuletić. Simulation of arbitrage-free implied volatility surfaces. *Appl. Math. Finance*, 30(2):94–121, 2023. doi: 10.1080/1350486X.2023.2277960.

Rama Cont, José da Fonseca, and Valdo Durrleman. Stochastic models of implied volatility surfaces. *Econ. Notes*, 31(2):361–377, 2002. doi: 10.1111/1468-0300.00090.

Jacopo Corbetta, Pierre Cohort, Ismail Laachir, and Claude Martini. Robust calibration and arbitrage-free interpolation of SSVI slices. *Decis. Econ. Finance*, 42(2):665–677, 2019.

Christa Cuchiero, Wahid Khosrawi, and Josef Teichmann. A generative adversarial network approach to calibration of local stochastic volatility models. *Risks*, 8(4):101, 2020. doi: 10.3390/risks8040101.

Petros Dellaportas and Aleksandar Mijatović. Arbitrage-free prediction of the implied volatility smile, 2014.

Fabrice Deschâtres. Convex volatility interpolation. SSRN working paper 4831218, 2024. URL https://papers.ssrn.com/abstract=4831218. Revised 2026.

Georgi Dimitroff, Dirk Röder, and Christian P. Fries. Volatility model calibration with convolutional neural networks. SSRN working paper 3252432, 2018. URL https://papers.ssrn.com/abstract=3252432.

Matthias R. Fengler. Arbitrage-free smoothing of the implied volatility surface. *Quant. Finance*, 9(4):417–428, 2009. doi: 10.1080/14697680802595585.

Matthias R. Fengler, Wolfgang K. Härdle, and Christophe Villa. The dynamics of implied volatilities: A common principal components approach. *Rev. Deriv. Res.*, 6(3):179–202, 2003. doi: 10.1023/B:REDR.0000004823.77464.2D.

Matthias R. Fengler, Wolfgang K. Härdle, and Enno Mammen. A semiparametric factor model for implied volatility surface dynamics. *J. Financial Econometrics*, 5(2):189–218, 2007. doi: 10.1093/jjfinec/nbm005.

Pascal François, Rémi Galarneau-Vincent, Geneviève Gauthier, and Frédéric Godin. Joint dynamics for the underlying asset and its implied volatility surface: A new methodology for option risk management. *J. Futures Mark.*, 46(3):545–561, 2026. doi: 10.1002/fut.70068.

Jim Gatheral. A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives. Presentation at Global Derivatives & Risk Management, Madrid, 2004.

Jim Gatheral and Antoine Jacquier. Arbitrage-free SVI volatility surfaces. *Quant. Finance*, 14 (1):59–71, 2014. doi: 10.1080/14697688.2013.819986.

Letitia Golubitsky, Matt Davison, and Alex Caraman. Unsupervised learning of equity market dynamics: A variational autoencoder approach. SSRN working paper 6726284, 2026. URL `https://papers.ssrn.com/abstract=6726284`.

Gaoyue Guo, Antoine Jacquier, Claude Martini, and Leo Neufcourt. Generalized arbitrage-free SVI volatility surfaces. *SIAM J. Financial Math.*, 7(1):619–641, 2016. doi: 10.1137/120900320.

Reinhold Hafner. *Stochastic Implied Volatility: A Factor-Based Model*, volume 545 of *Lecture Notes in Economics and Mathematical Systems*. Springer Berlin Heidelberg, 2004. ISBN 978-3-540-22183-8. doi: 10.1007/978-3-642-17117-8. URL `https://books.google.com/books?id=67OE8cKXWBQC`.

Yinbin Han, Jack Yuxiang Zhang, Manuel Torres, Fernando Acero, and Renyuan Xu. Diffusion models for dynamic volatility surface generation and data-driven hedging, 2026. URL `https://arxiv.org/abs/2609.13402`.

Lifeng Hao and Shaolin Ji. Decoupled probabilistic forecasting and arbitrage-aware refinement of implied volatility surfaces, 2026. URL `https://arxiv.org/abs/2607.29220`.

Sebas Hendriks and Claude Martini. The extended SSVI volatility surface. *J. Comput. Finance*, 22(5):25–39, 2019. doi: 10.21314/JCF.2019.365.

Andres Hernandez. Model calibration with neural networks. SSRN working paper 2812140, 2016. URL `https://papers.ssrn.com/abstract=2812140`.

Blanka Horvath, Aitor Muguruza, and Mehdi Tomas. Deep learning volatility, 2019. URL `https://arxiv.org/abs/1901.09647`.

Andrey Itkin. Deep learning calibration of option pricing models: Some pitfalls and solutions, 2019. URL `https://arxiv.org/abs/1906.03507`.

Chen Jin and Ankush Agarwal. Forecasting implied volatility surface with generative diffusion models, 2025.

Nabil Kahalé. An arbitrage-free interpolation of volatilities. *Risk*, 17(5):102–106, 2004.

Jan Kallsen and Paul Krühner. On a Heath–Jarrow–Morton approach for stock options. *Finance Stoch.*, 19(3):583–615, 2015. doi: 10.1007/s00780-015-0263-1.

Shuaiqiang Liu, Anastasia Borovykh, Lech A. Grzelak, and Cornelis W. Oosterlee. A neural network-based framework for financial model calibration. *J. Math. Ind.*, 9:9, 2019. doi: 10.1186/s13362-019-0066-7.

Claude Martini and Arianna Mingone. No arbitrage SVI. *SIAM J. Financial Math.*, 13(1): 227–261, 2022.

Nikita Medvedev and Zhiguang Wang. Multistep forecast of the implied volatility surface using deep learning. *J. Futures Mark.*, 42(4):645–667, 2022. doi: 10.1002/fut.22302.

Arianna Mingone. No arbitrage global parametrization for the eSSVI volatility surface. *Quant. Finance*, 22(12):2205–2217, 2022.

Brian Ning, Sebastian Jaimungal, Xiaorong Zhang, and Maxime Bergeron. Arbitrage-free implied volatility surface generation with variational autoencoders. *SIAM J. Financial Math.*, 14(4): 1004–1027, 2023. doi: 10.1137/21M1443546.

Gergely Orosi. Arbitrage-free call option surface construction using regression splines. *Appl. Stoch. Models Bus. Ind.*, 31(4):515–527, 2015. doi: 10.1002/asmb.2045.

Nikolaos Panigirtzoglou and George Skiadopoulos. A new approach to modeling the dynamics of implied distributions: Theory and evidence from the S&P 500 options. *J. Bank. Finance*, 28 (7):1499–1520, 2004. doi: 10.1016/S0378-4266(03)00127-4.

Philip E. Protter. *Stochastic Integration and Differential Equations*. Springer, second edition, 2005.

Hao Qin, Ruozhong Yang, Charlie Y. Che, and Liming Feng. Arbitrage-free multi-maturity risk-neutral marginals. arXiv preprint arXiv:2607.06204, 2026. URL `https://arxiv.org/abs/2607.06204`.

Dirk Roeder and Georgi Dimitroff. Volatility model calibration with neural networks: A comparison between direct and indirect methods, 2020. URL `https://arxiv.org/abs/2007.03494`.

Thomas Roos. Simple, flexible, analytic, arbitrage free option price interpolation. SSRN working paper 5215592, 2025. URL `https://papers.ssrn.com/abstract=5215592`. Revised 2026.

Mathieu Rosenbaum and Jianfei Zhang. Deep calibration of the quadratic rough Heston model, 2021. URL `https://arxiv.org/abs/2107.01611`.

Sebastian Schlenkrich. A piece-wise model for vanilla option pricing. SSRN working paper 5214566, 2025. URL `https://papers.ssrn.com/abstract=5214566`.

Philipp J. Schönbucher. A market model for stochastic implied volatility. *Philos. Trans. R. Soc. Lond. A*, 357(1758):2071–2092, 1999. doi: 10.1098/rsta.1999.0418.

Martin Schweizer and Johannes Wissel. Arbitrage-free market models for option prices: The multi-strike case. *Finance Stoch.*, 12(4):469–505, 2008a. doi: 10.1007/s00780-008-0068-6.

Martin Schweizer and Johannes Wissel. Term structures of implied volatilities: Absence of arbitrage and existence results. *Math. Finance*, 18(1):77–114, 2008b. doi: 10.1111/j.1467-9965.2007.00323.x.

George S. Skiadopoulos, Stewart D. Hodges, and Les Clewlow. The dynamics of the S&P 500 implied volatility surface. *Rev. Deriv. Res.*, 3(3):263–282, 2000. doi: 10.1023/A:1009642705121.

Abir Sridi and Paul Bilokon. Applying deep learning to calibrate stochastic volatility models, 2023. URL `https://arxiv.org/abs/2309.07843`.

Milena Vuletić and Rama Cont. VolGAN: A generative model for arbitrage-free implied volatility surfaces. *Appl. Math. Finance*, 31(4):203–238, 2024. doi: 10.1080/1350486X.2025.2471317.

Ruben Wiedemann, Antoine Jacquier, and Lukas Gonon. Operator deep smoothing for implied volatility. In *International Conference on Learning Representations*, 2025. URL `https://arxiv.org/abs/2406.11520`.

Magnus Wiese, Lianjun Bai, Ben Wood, and Hans Buehler. Deep hedging: Learning to simulate equity option markets, 2019. URL `https://arxiv.org/abs/1911.01700`.

Magnus Wiese, Ben Wood, Alexandre Pachoud, Ralf Korn, Hans Buehler, Phillip Murray, and Lianjun Bai. Multi-asset spot and option market simulation, 2021. URL `https://arxiv.org/abs/2112.06823`.

Johannes Wissel. Arbitrage-free market models for option prices. NCCR FINRISK Working Paper 428, NCCR FINRISK, ETH Zürich, 2007.

Mingzhi Yang, Sheng Wang, Chao Zhang, and Ruikun Li. Universal diffusion models for implied volatility surfaces: Learning shared dynamics across stocks, 2026.

Yongxin Yang, Wenqi Chen, Chao Shu, and Timothy M. Hospedales. HyperIV: Real-time implied volatility smoothing. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 70550–70564. PMLR, 2025. URL `https://proceedings.mlr.press/v267/yang25d.html`.

Chen Zhang, Giovanni Amici, and Marco Morandotti. Calibrating the Heston model with deep differential networks, 2024. URL `https://arxiv.org/abs/2407.15536`.

Wenyong Zhang, Lingfei Li, and Gongqiu Zhang. A two-step framework for arbitrage-free prediction of the implied volatility surface. *Quant. Finance*, 23(1):21–34, 2023. doi: 10.1080/14697688.2022.2135454.